家禽腫瘤性疾病

Neoplastic Diseases of Poultry

吳乃慧 (Nai-Huei Wu) DVM, PhD

Neoplastic diseases

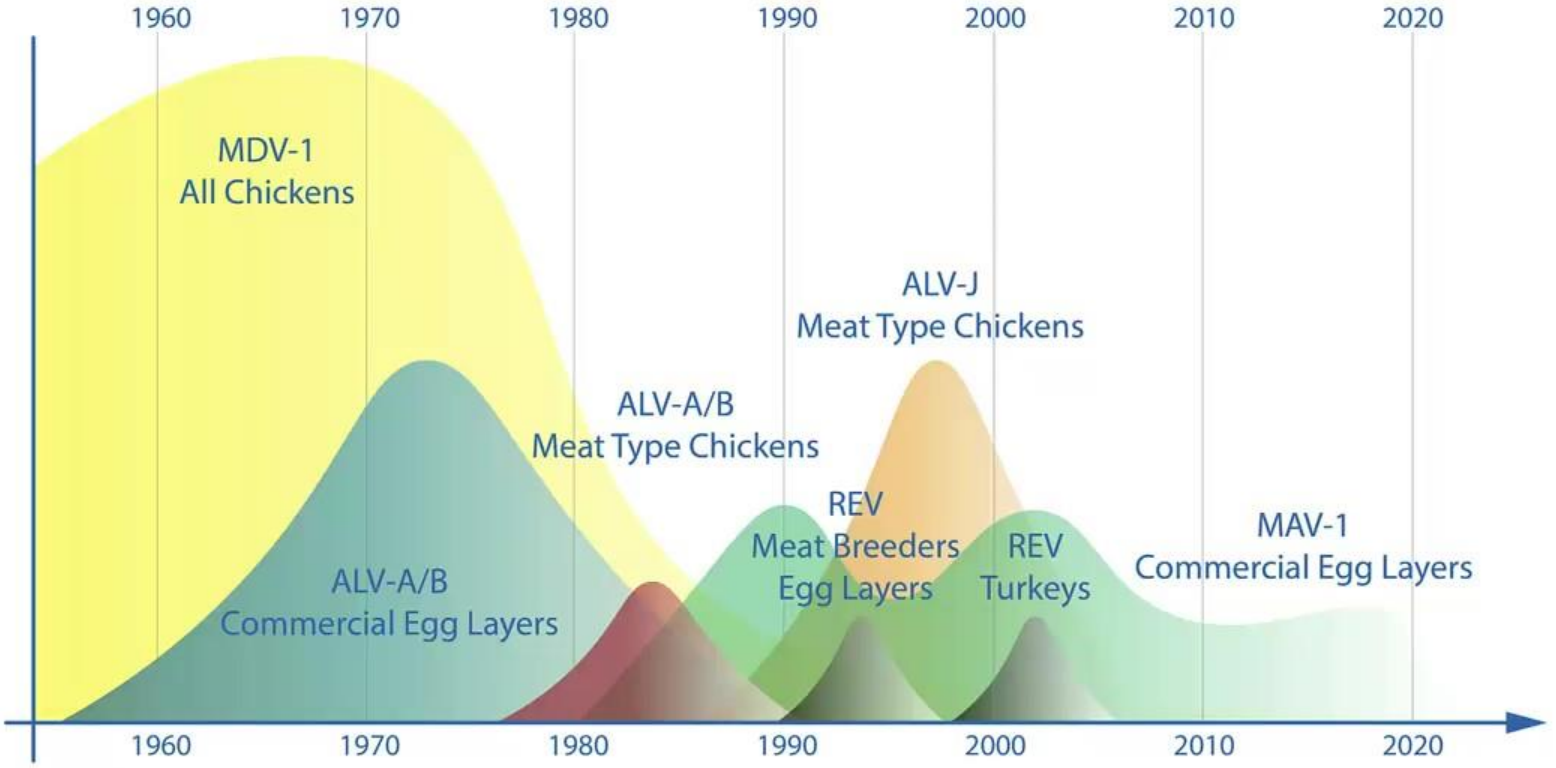
- infectious
  - economic importance
  - widely prevalent in commercial stock
  - affect relatively young birds
- non-infectious
  - no great economic significance
  - sporadic
  - older than the usual lifespan of commercial birds

Virus-induced transmissible neoplastic diseases

- Marek's disease 馬立克病
  - Marek's disease virus (MDV), herpesvirus
- Avian leukoses/sarcoma 家禽白血病
  - avian leukosis virus (ALV), retrovirus
- Reticuloendotheliosis 網狀內皮細胞增殖病
  - reticuloendotheliosis virus (REV), retrovirus

Virus type	Nucleic acid type	Virus classification of etiological agent	Neoplastic diseases
Retrovirus	RNA	Leukosis/sarcoma group	Leukoses Lymphoid leukosis Erythroblastosis Myeloloblastosis Sarcomas and other connective tissue tumors Fibrosarcoma, fibroma Myxosarcoma, myxoma Osteogenic sarcoma, osteoma Histiocytic sarcoma Related neoplasms Hemangioma Nephroblastoma Hepatocarcinoma Osteopetrosis
		Reticuloendotheliosis group	Reticuloendotheliosis Lymphoid leukosis
Herpesvirus	DNA	Marek's disease virus	Marek's disease

Historical Perspective



G. Zavala. AAAP2021

Marek's Disease

馬立克氏病

吳乃慧 (NAI-HUEI WU)

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OUTLOOK

- **Marek's disease**
  - **Epidemiology**
  - **Diagnosis**
  - **Control**
  - **Taiwan**

HISTORY



- 1907 匈牙利 獸醫/病理學家 Prof. József Marek
  - University of Veterinary Medicine Budapest
  - Clinical symptom: Polyneuritis 周邊神經病變

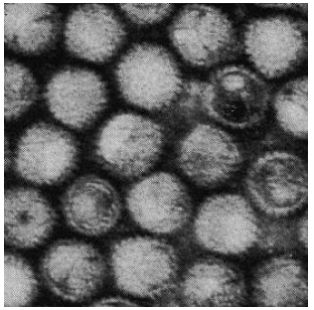




- 1926 - Neoplastic diseases in visceral organs

HISTORY

- 1950' s – 美國稱之為 Visceral lymphomatosis / acute leucosis
- 1967 英國 Dr. P.M. Biggs 證實 Alpha Herpesvirus



- 1970 起, 在美國 每年2百萬美金的損失
- 直至目前, 每年因Marek's Disease 所損失的金額依舊很高

Impact of Marek's Disease

Economic Impact	Year	Region	Reference
≥ $943 million	1985	World	Purchase H.G. Marek's Disease, Scientific basis and methods of control. Ed by L.N. Payne. Martinus Nijhoff Publishing, Boston Mass, 1985 p17-42.
≥ $150 million	2004	USA	Fadly A.M. Proc. XII Moscow Int Congress Small Animal Diseases, p26, 2004.
$ 1-2 billion	2004	World	Morrow C. and F. Fehler. Marek's Disease: a Worldwide Problem. In: Marek's Disease: an evolving problem. T.F. Davison and V.K. Nair. eds. Elsevier Academic Press, London, England pp 49-61, 2004.

G. Zavala. AAAP2021

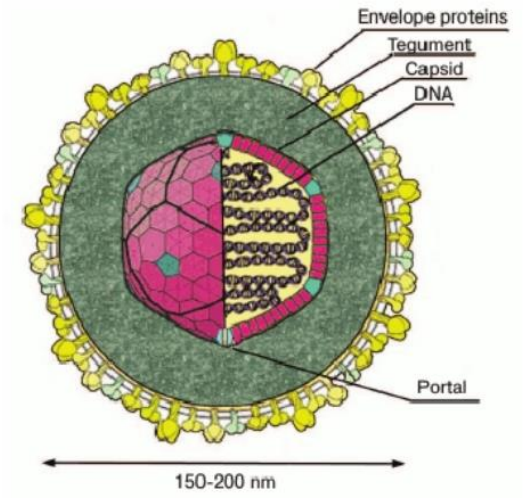
MD VIRUS

- 科: Herpesviridae
- 屬: Mardivirus
- 種: Gallid herpesvirus 2-3 (GaHV)
- 血清型 3 Serotypes

Infectious Laryngotracheitis (ILT)

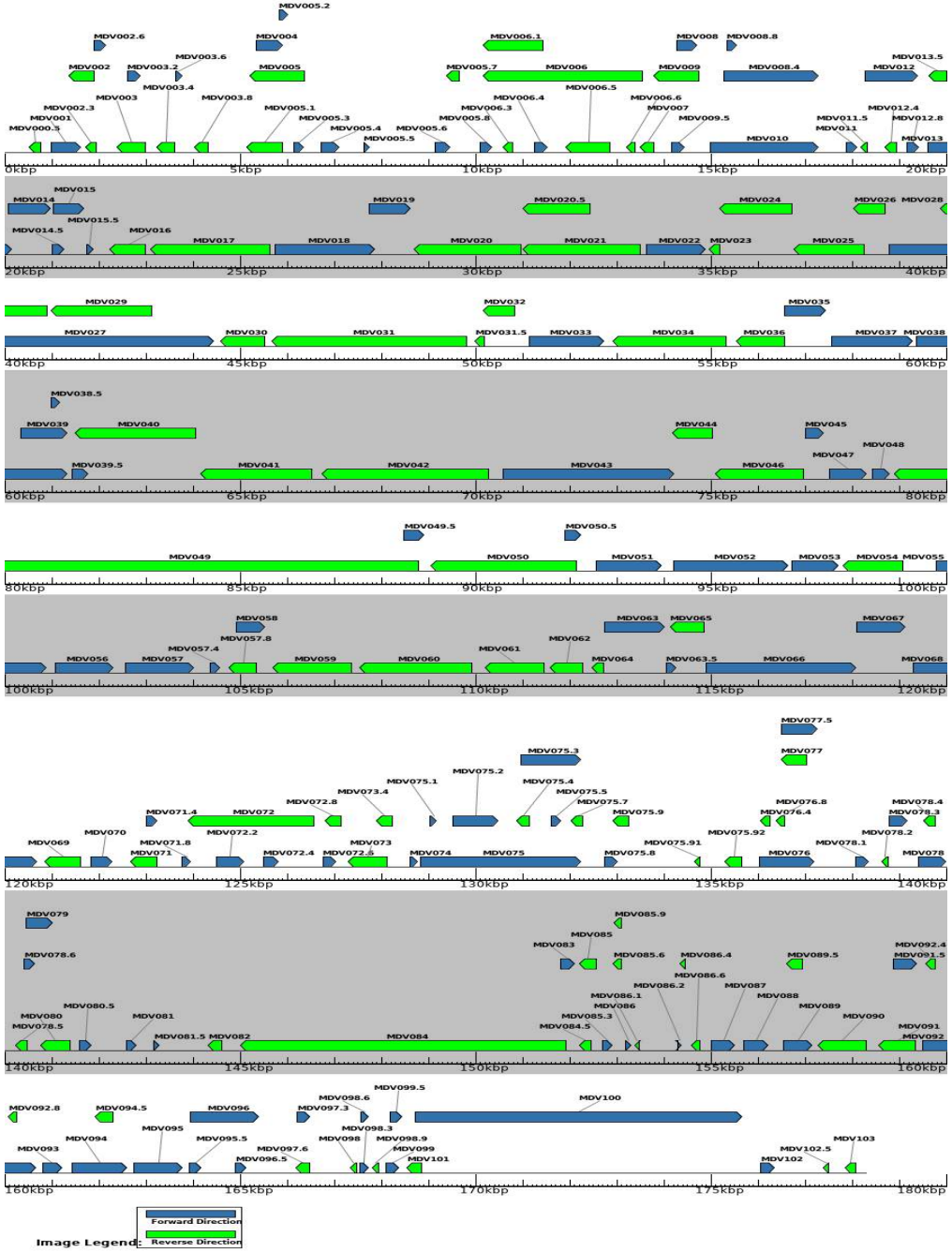
Order	Family	Sub-family	Genus	Species
Herpesvirales	*Alloherpesviridae*		*4 Genera*	
	Malacoherpesviridae		*2 Genera*	
	Herpesviridae	*Alphaherpesvirinae*	*Iltovirus*	*Gallid herpesvirus 1*
				Psittacid herpesvirus 1
			Mardivirus	*Anatid herpesvirus 1*
				Columbid herpesvirus 1
				Gallid herpesvirus 2 (MDV-1)
				Gallid herpesvirus 3 (MDV-2)
				Meleagrid herpesvirus 1 (MDV-3, HVT)
			Scutavirus	*Chelonid herpesvirus 5*
			Simplexvirus	*11 species*
			Varicellovirus	*17 Species*
		Betaherpesvirinae	*4 Genera*	
		Gammaherpesvirinae	*4 Genera*	
		Unassigned	*Unassigned*	*Iguanid herpesvirus 2*

Epidemiology

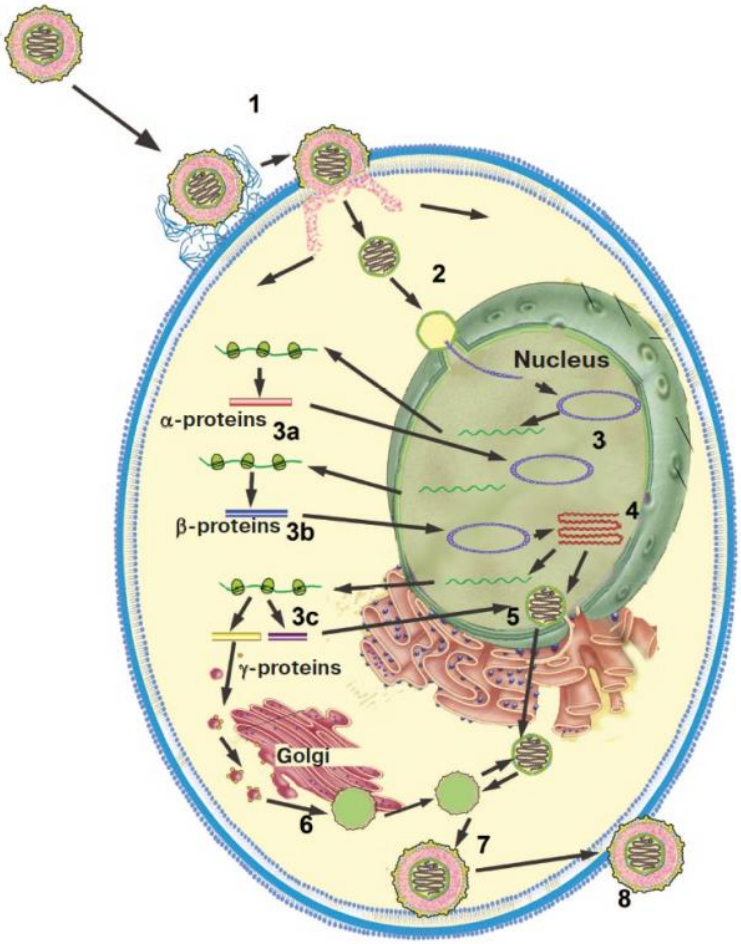


- 20面/球體
- 直徑150-200nm
- Enveloped DNA virus
- ds-DNA, Linear 175kb
- ~190 proteins

Sequence of CVI988



VIRUS REPLICATION CYCLE



1. Attach by Envelop
2. Uncoating capsuled dsDNA
3. viral gene expression:
   - 3a immediate early
   - 3b early
   - 3c late genes.
4. Nucleocapsid assembly
5. Nuclear (dsDNA) egress
6. Capsule maturation
7. Assembling
8. Budding

SEROTYPES

	Type I	Type II	Type III
	Gallid herpesvirus type 2	Gallid herpesvirus type 3	Meleagrid herpesvirus type 1
Pathogenesis	Oncogenic	Non-oncogenic	Non-oncogenic
Vaccine	**CVI988**/Rispens	**SB-1**	**HVT**
Virulent	vv+MDV, vvMDV, vMDV	mMDV avirulent	Non



Very Virulent (+plus) mild

HOSTS

- Mainly domestic **Chickens**
- some Turkeys / Quails
- Migratory waterfowl in Japan

- Can be infected by more than 1 MDV strain.
- 各品種的雞感受性有差異 (產生腫瘤機率不同)

SPREAD

- **Direct** / indirect contact – Airborne route

- Infectious virus in the **Feather follicle epithelium (FFE)** (羽毛毛囊細胞)→ contaminate the environment

- chicken infected by MDV – 2~4wk virus shedding – Clinical symptoms

- latent infection (終身帶毒)

- 年紀越小接觸病毒發病率越高

- Multifactor – female showed more tumours than males.

CLINICAL SIGNS

感染病毒後至少需4週症狀才會出現(潛伏期)，自然發病通常在6週齡後→種雞、蛋雞、有色雞

	Mortality		**Main symptoms**	
Classical (Chronic)	< 10-15%		Paralysis of legs	Neural type Ocular type
Acute	Intermediate-High 迅速增加	＞vMDV	General manifestation Visceral tumor	發生率10-30%， Up to 70%
Acute cytolytic	Very High Dead: 10-14d (age)	＞**vvMDV**	Atrophy of lymphoid Tumor	Early mortality syndrome
Transient paralysis	Low Occur 5-18wks	mMDV	Ataxia(共濟失調), paralysis	Uncommon

MARCOSCOPIC LESIONS 4 TYPES OF MDV

1. 神經型	Neural form	(chronic)	– 外觀/ 剖檢
2. 內臟型	Visceral form	(acute)	– 剖檢
3. 皮膚型	Skin form		– 外觀
4. 眼型	Ocularal form	(chronic)	– 外觀

(1) 神經型 (NEURAL FORM)



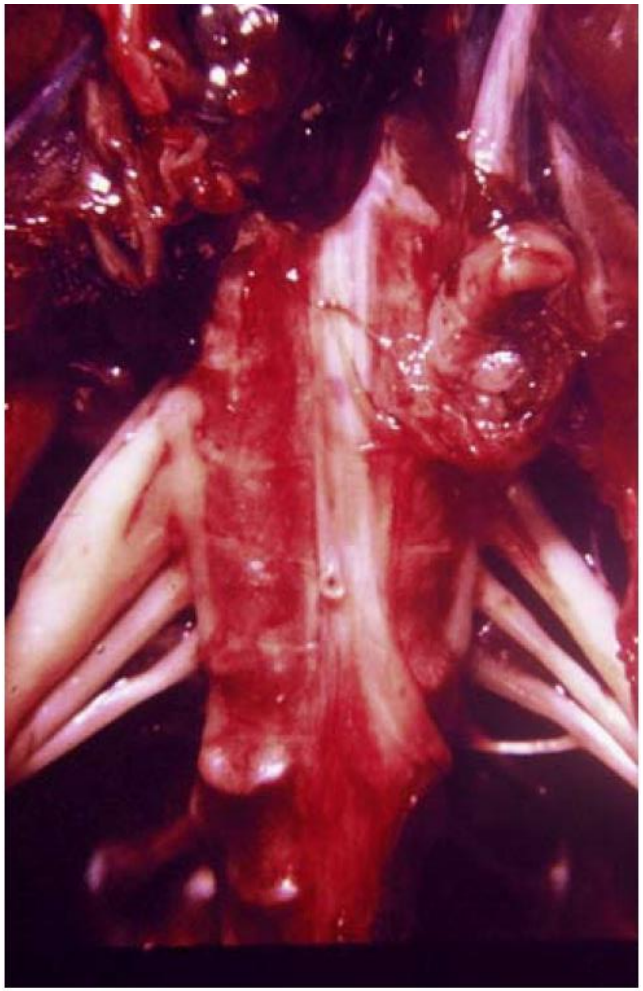
- Paralysis of limbs





NEURAL FORM





Sciatic nerve 坐骨神經

- **Unilateral**/ bilateral thickening of affected nerves
- 神經腫大、失去光澤呈暗灰色

NEURAL FORM

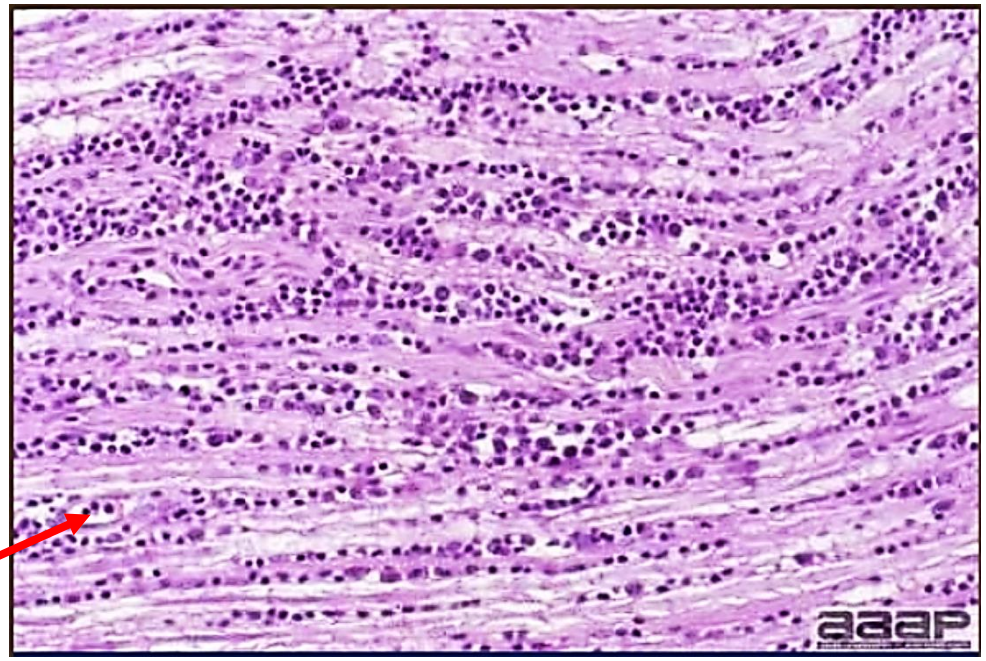
Neural form:

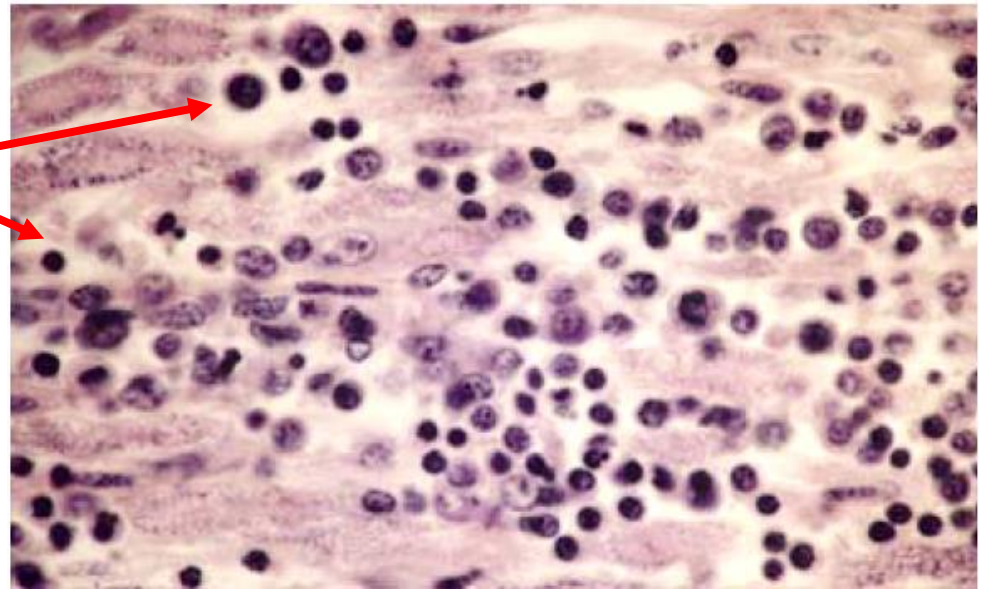
In Peripheral nerves:

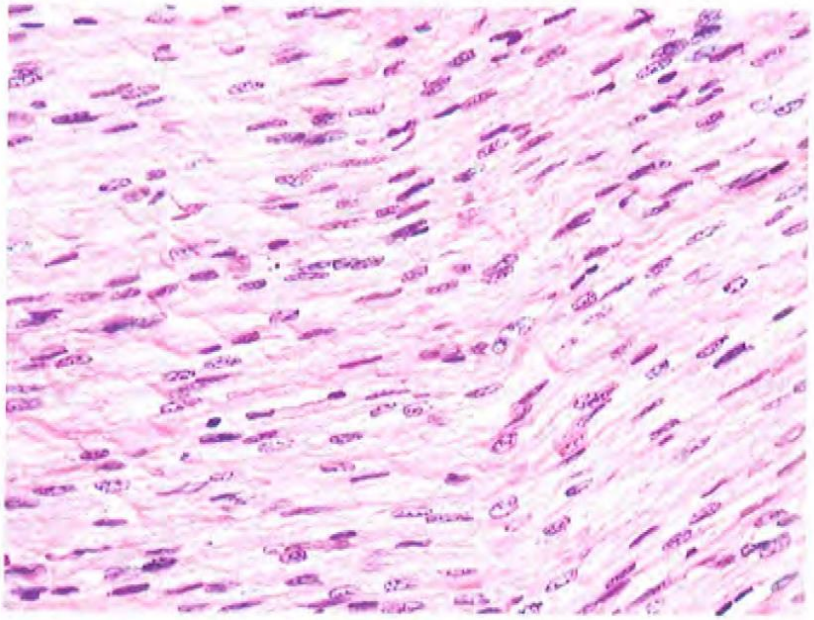
(a) Mature **lymphocytes**

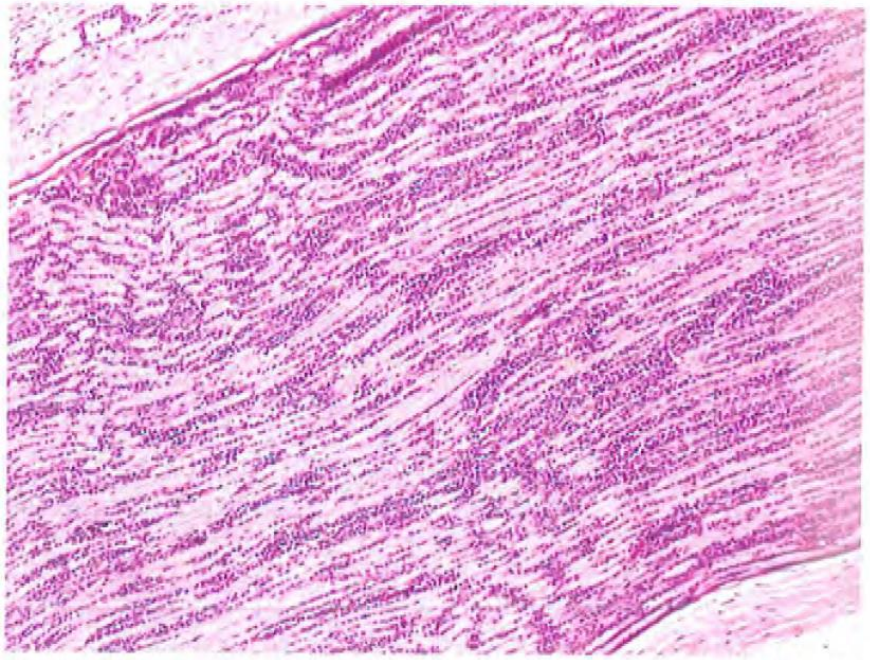
(b) edema

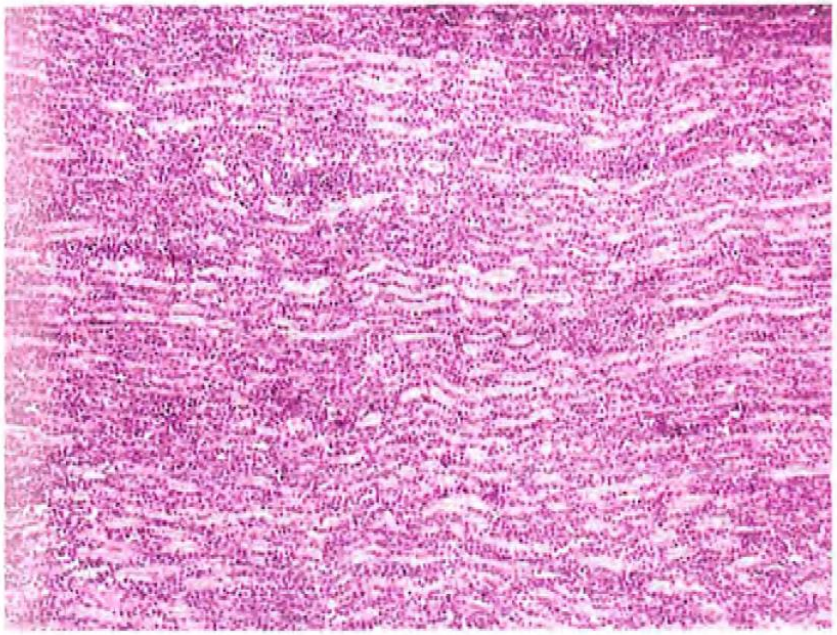
(c)大小不一淋巴球

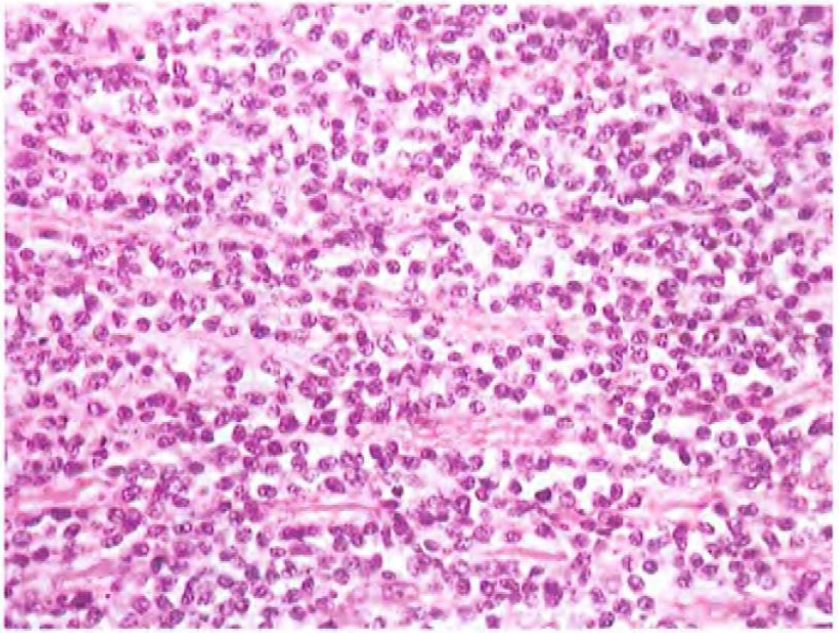


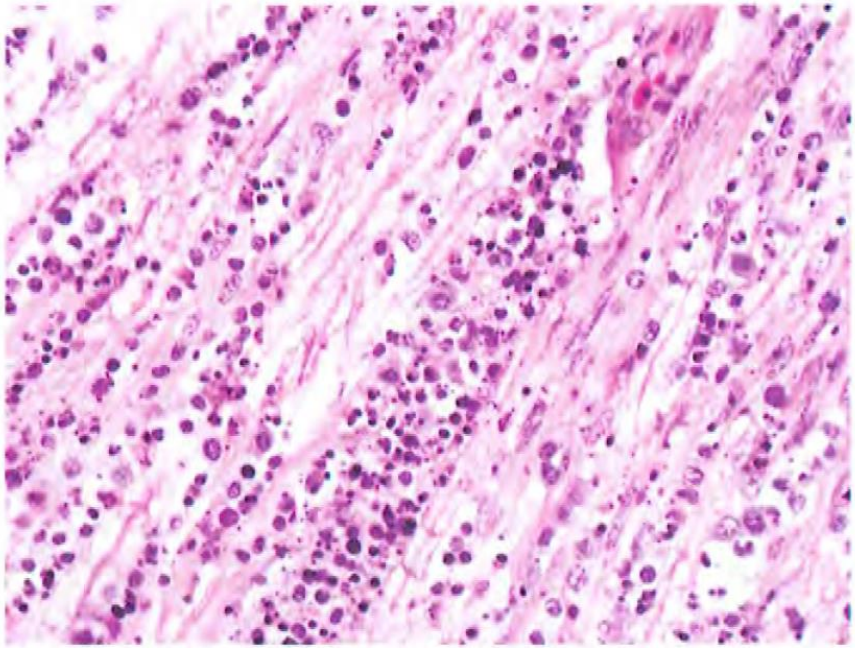












(2)内臓型 (VISCERAL FORM)

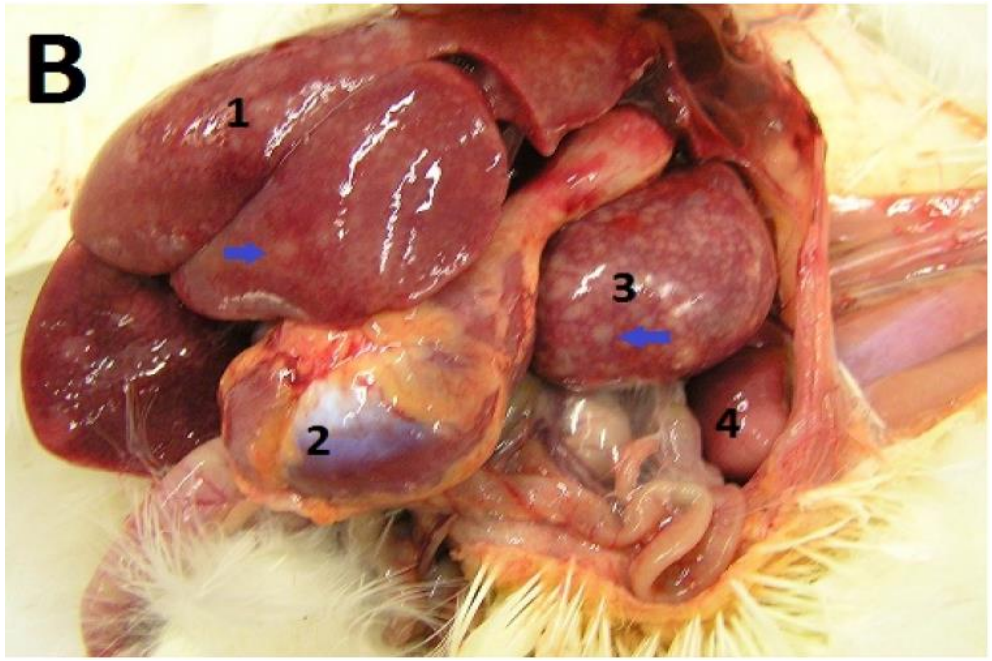
Visceral form:

- widespread, diffuse lymphomatous involvement of visceral organs

- proventriculus, liver, spleen,

- ovary, kidney,

- heart, others

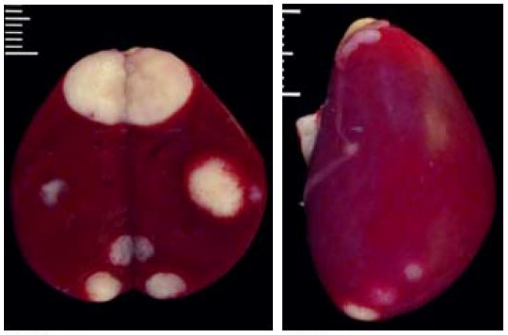


VISCERAL FORM



Spleen





HJ Barnes HJ Barnes

- various tumor nodules form in liver
- Spleen enlargement

VISCERAL FORM





Sanders



- Proventriculus
- Shape of a round bottom flask
- diffuse neoplastic growth, narrow lumen.





VISCERAL FORM



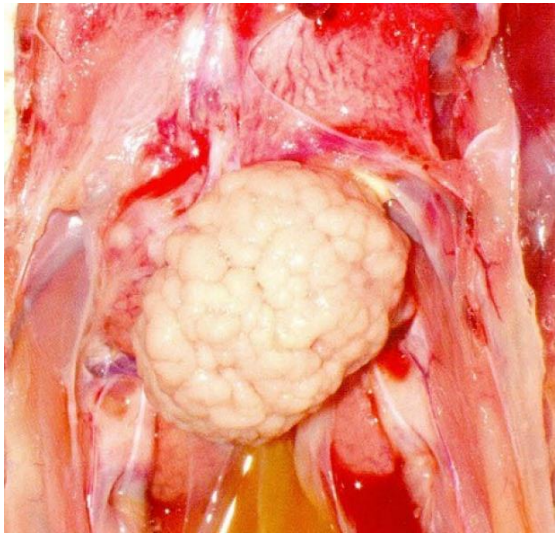
- Pancreas
- diffuse neoplastic growths



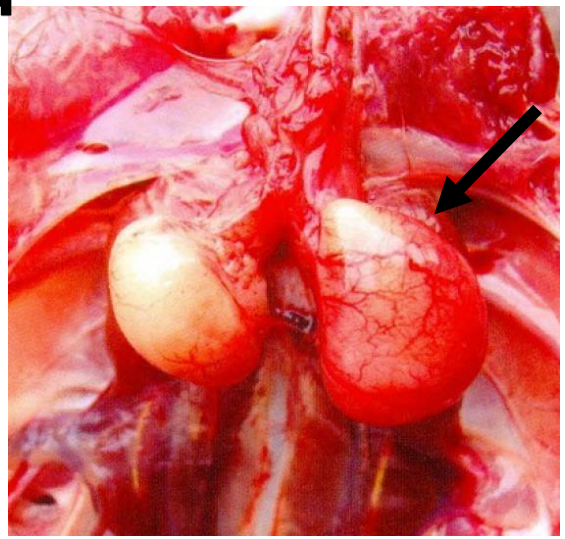
HJ Barnes

VISCERAL FORM

♀

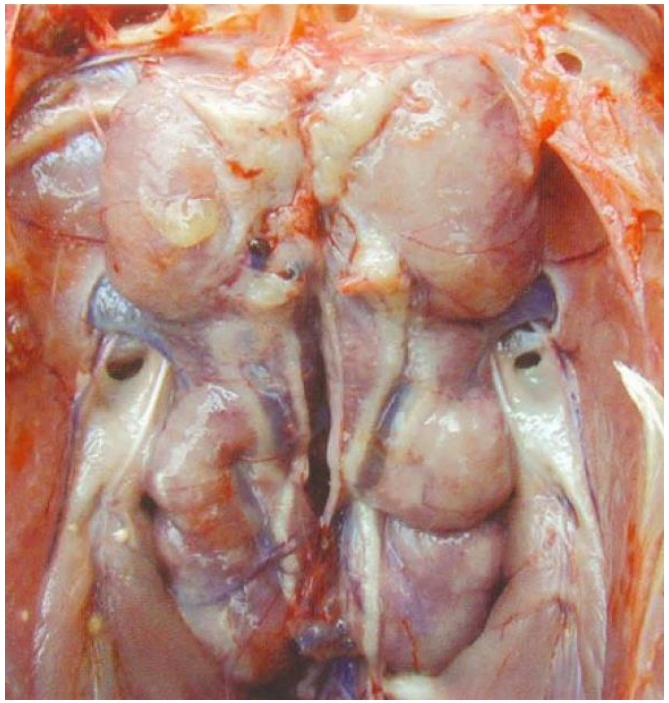


♂

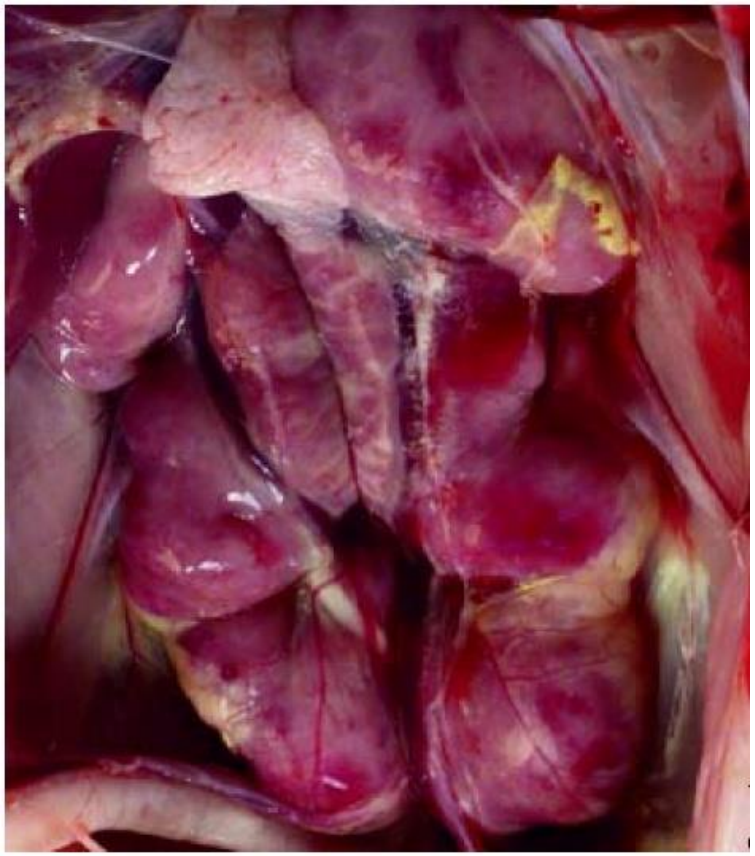


- Typical cauliflower-like ovary (花椰菜狀)
- Asymmetry of testes, unilateral lymphoid cell proliferation

VISCERAL FORM



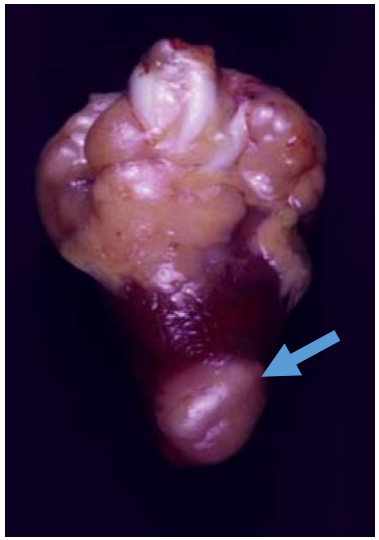
Kidney

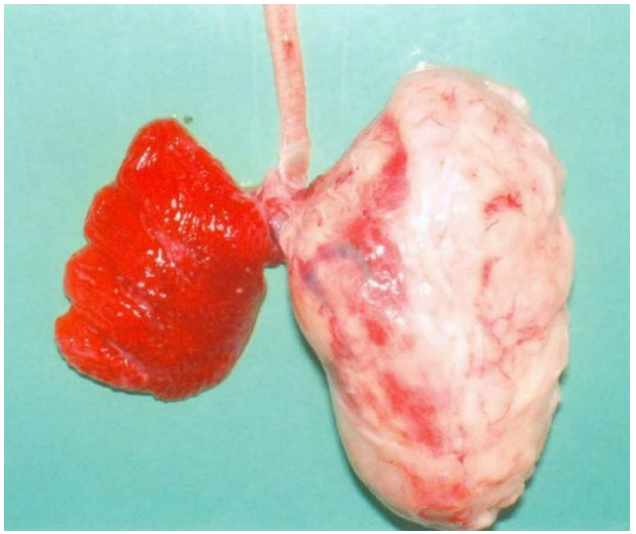


- Enlargement of kidneys - Bilateral

VISCERAL FORM

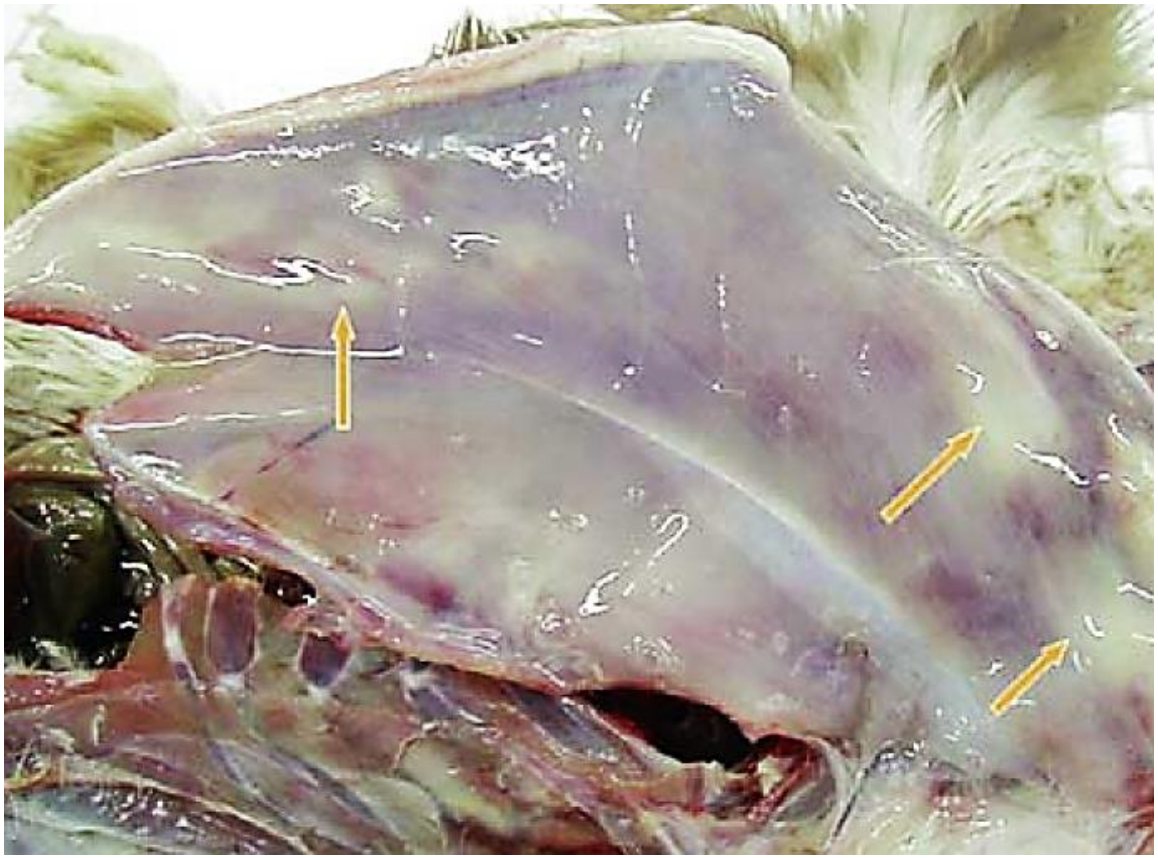






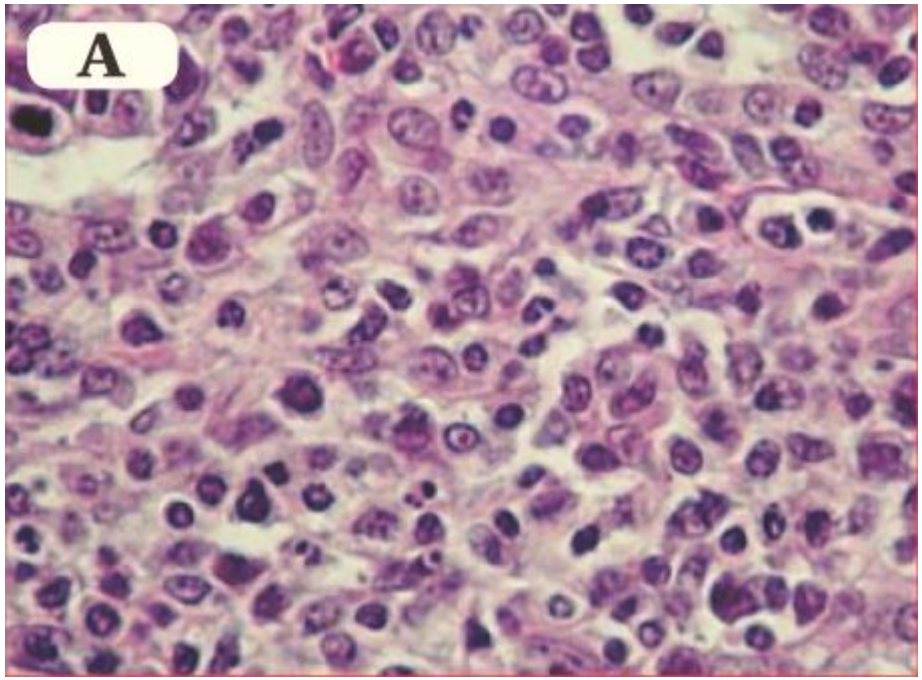
- Diffuse lymphomatous growths in heart
- Neoplastically modified right lung.

VISCERAL FORM



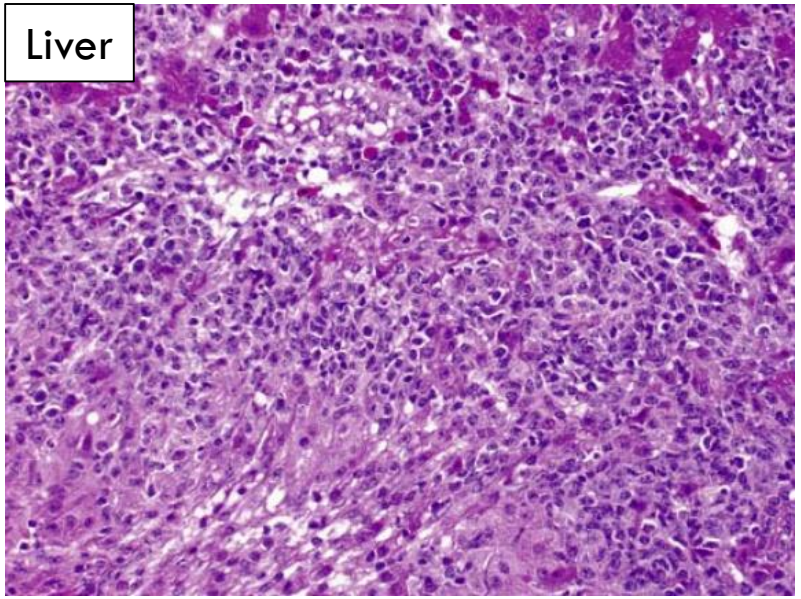
- multicentral MD tumors in pectoral muscles

VISCERAL FORM

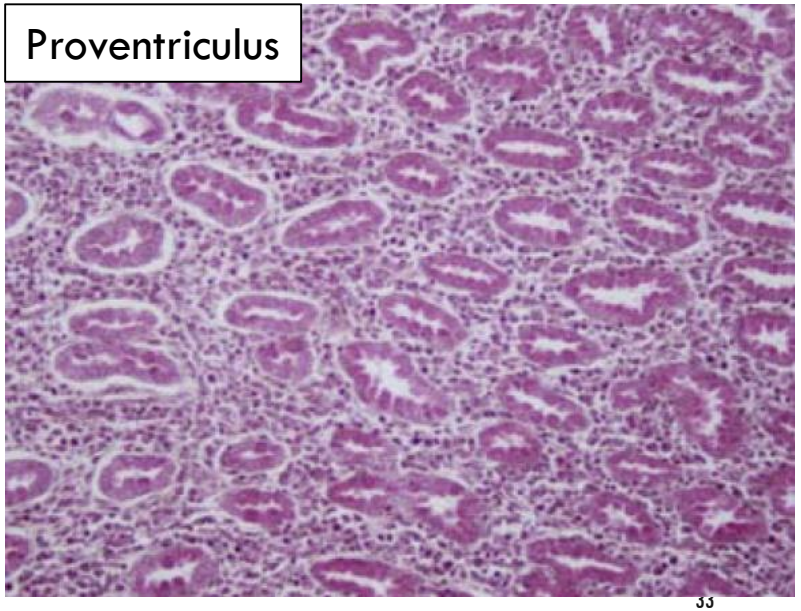


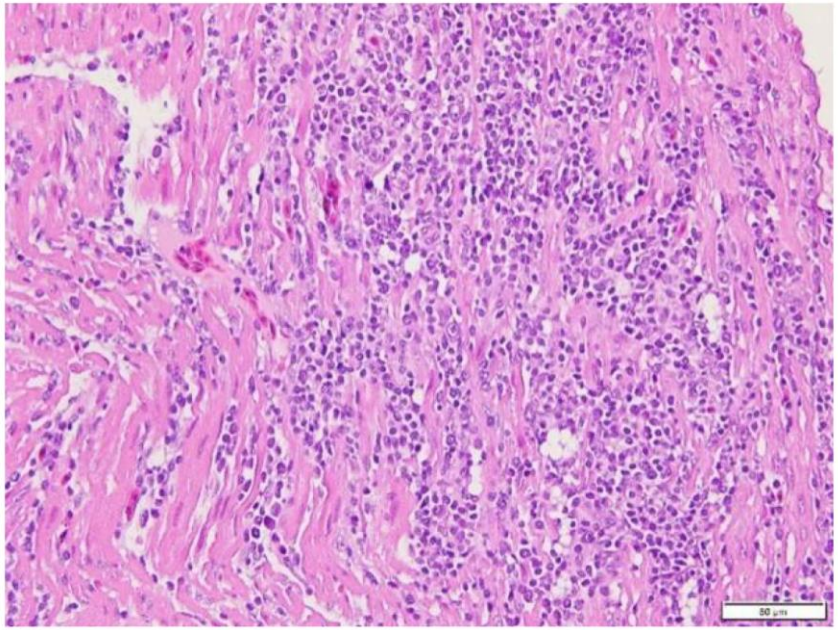
- Histologically, pleiomorphic lymphoid cell proliferation.

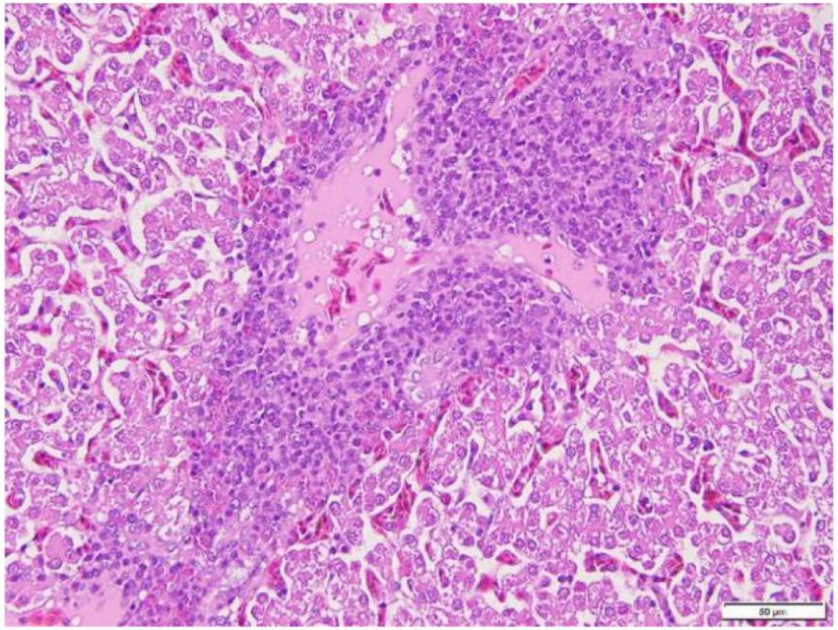
Liver



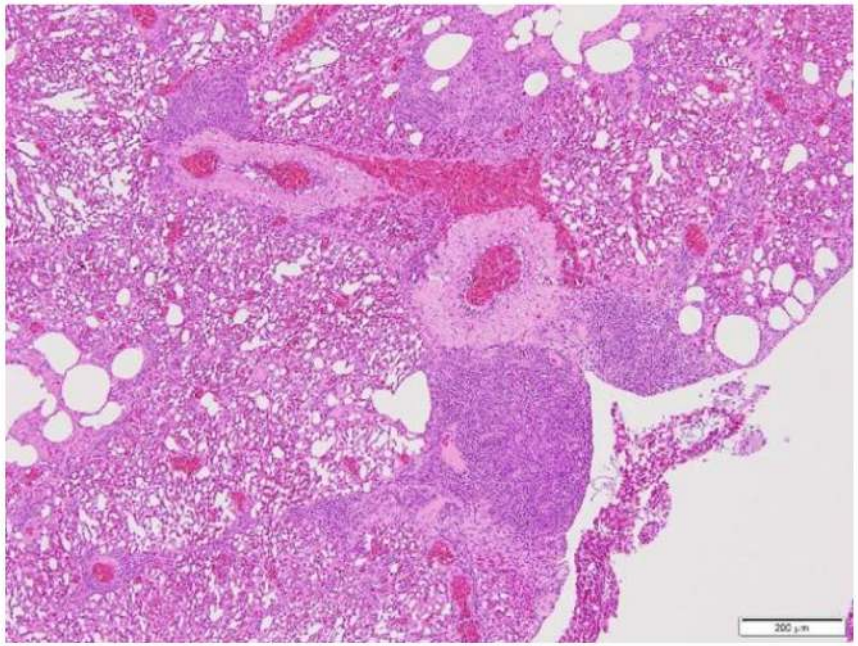
Proventriculus

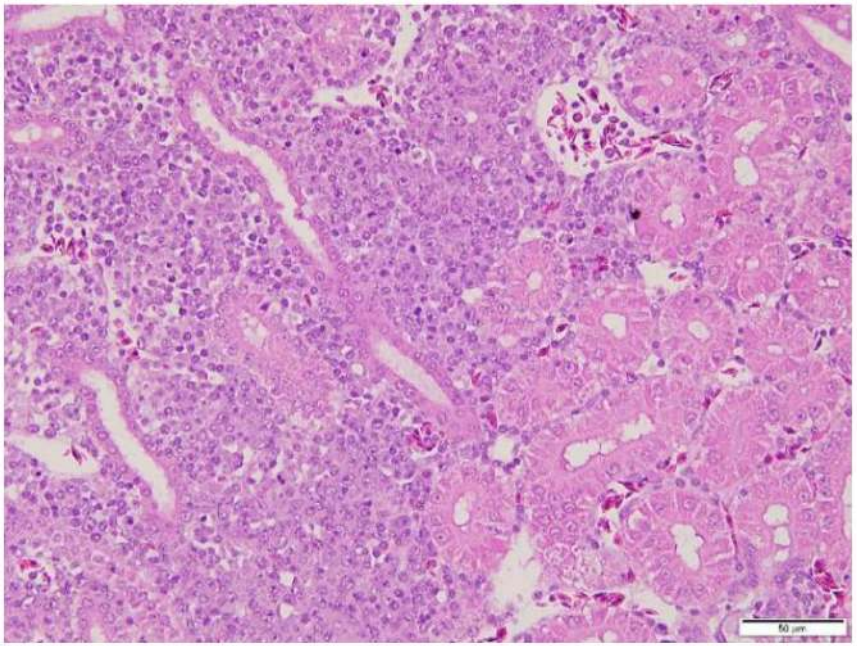






(郭,2018)





(郭,2018)

(3)皮膚型 (CUTANEOUS FORM)



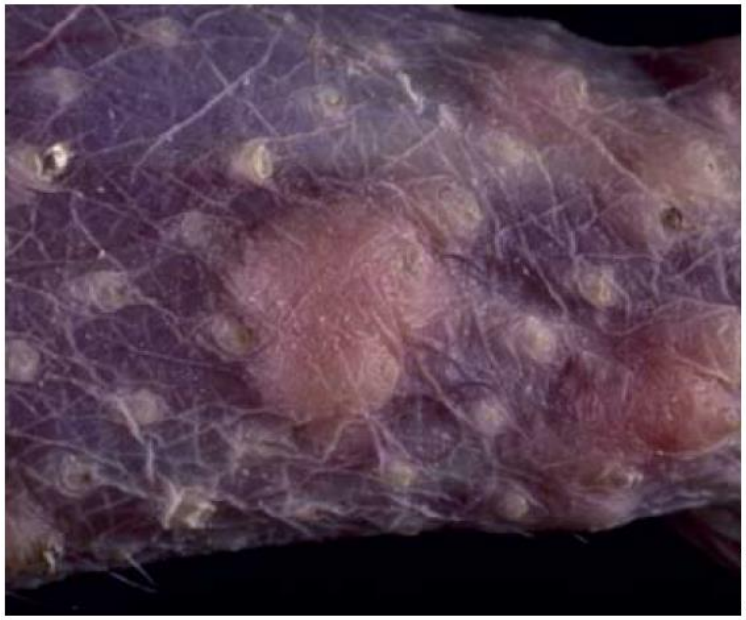




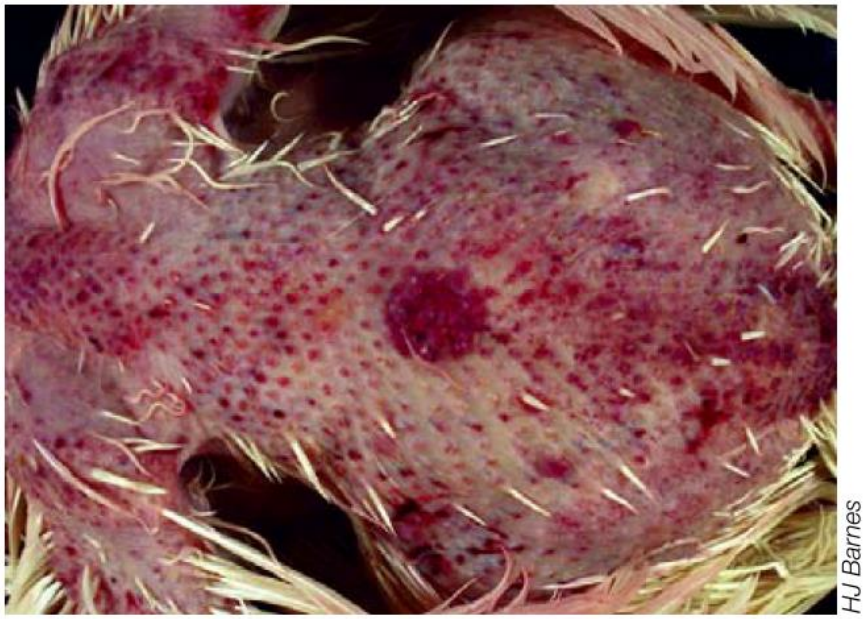
Enlarged feather follicles

- Cutaneous form
- Enlargement of feather follicles

CUTANEOUS FORM



Sanders

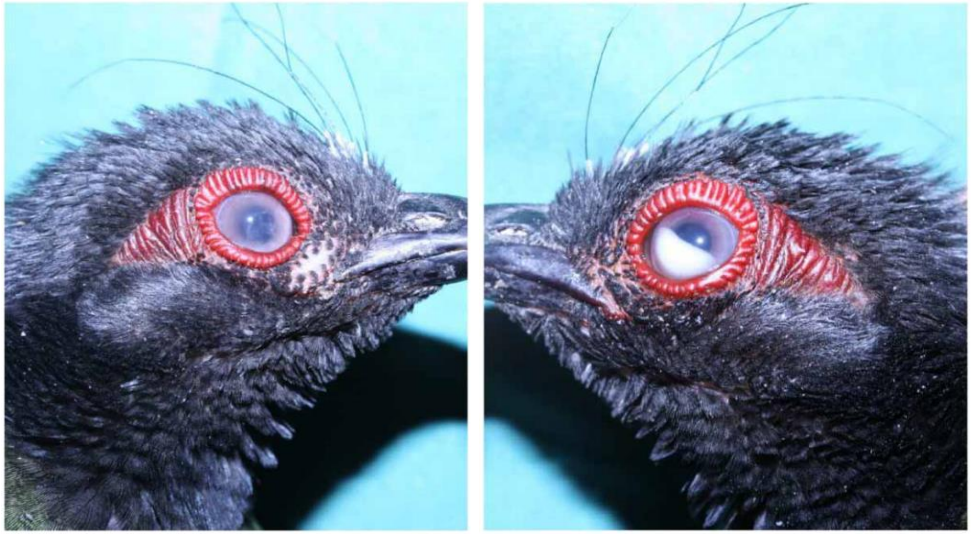


HJ Barnes

- Tumors of feather follicles after plucking (in slaughterhouse)
- Nodular lesions, scattered follicles may coalesce, reddening of the skin

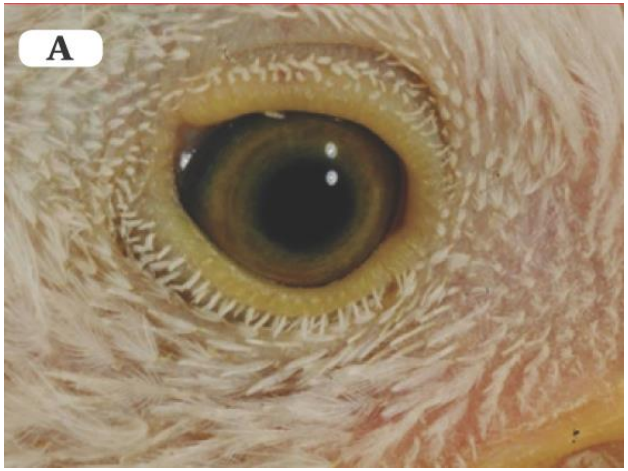
(4) 眼型 (OCULAR FORM)

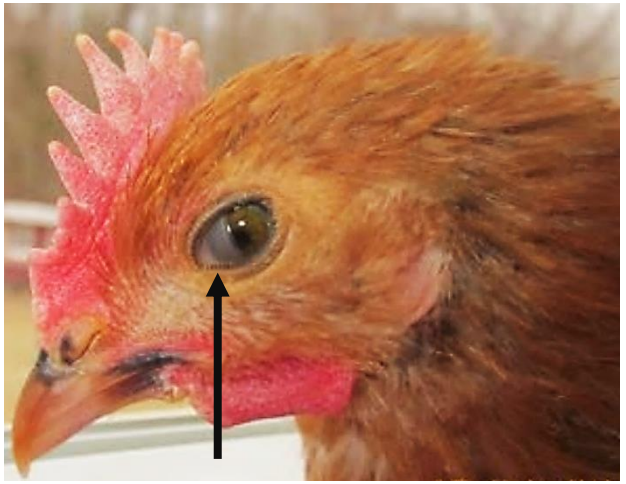
Abnormal corneal/ Iris



- Ocular form, chronic/classical form.
- Iris depigmentation
- deformation of pupil
- Opacity of cornea / blindness

normal



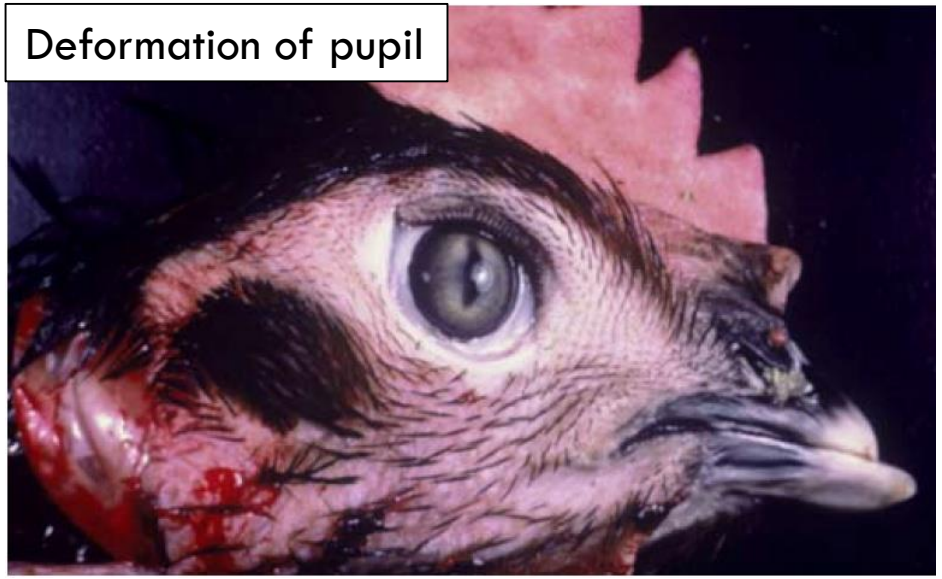


Nictitating membrane(瞬膜)

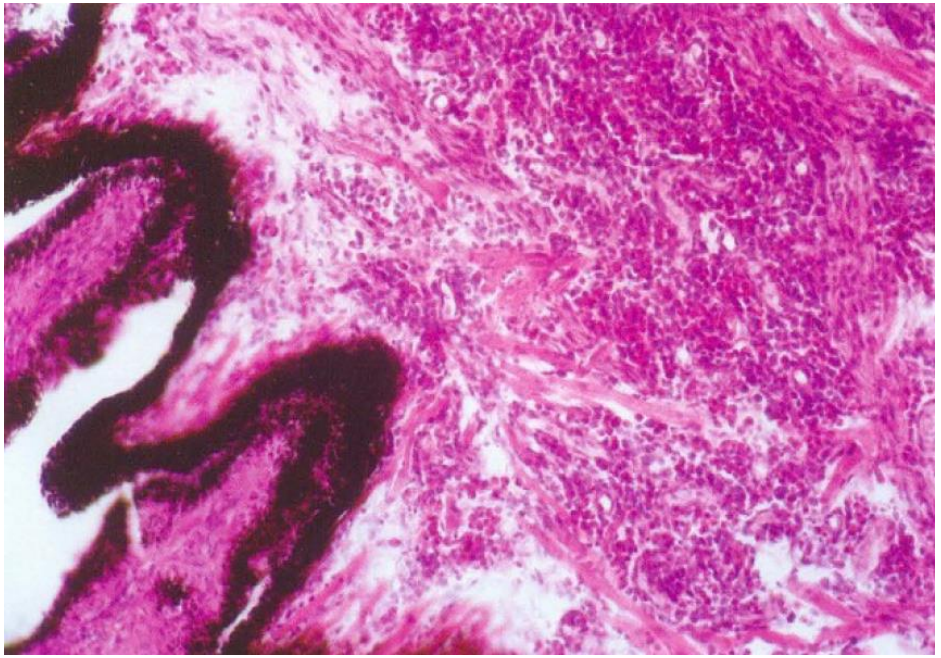
瞳孔變形

OCULAR FORM

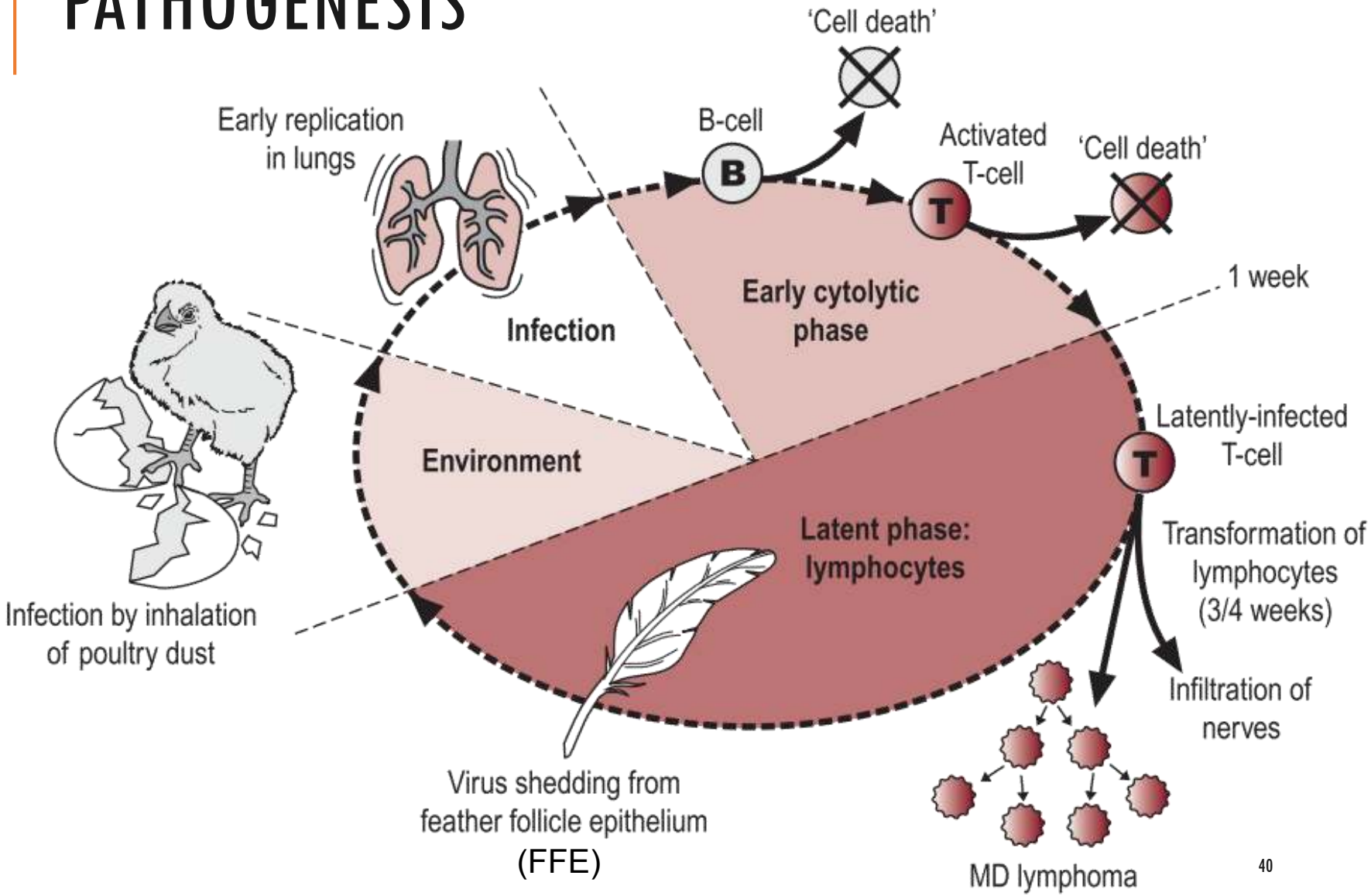
- Vaccines cannot fully protect the Ocular form of MDV infection.



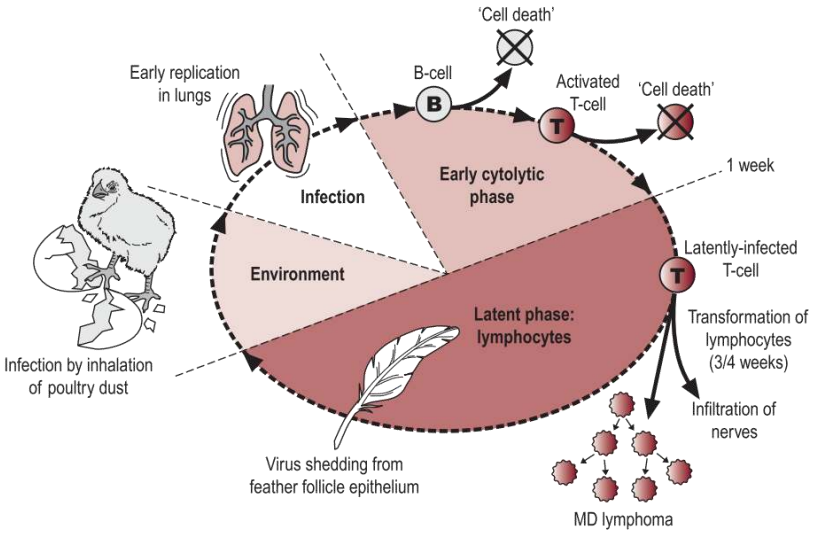
- Lymphoid cell proliferations in the iris and the muscles



PATHOGENESIS



PATHOGENESIS



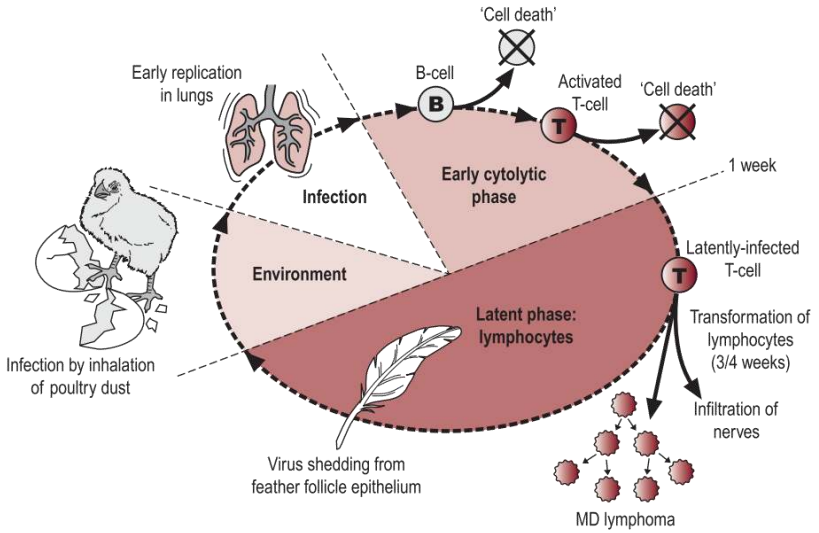
1. Contaminated environment
2. Chicks inhalation
3. Uptake by lung “phagocyte”
4. Infection of B cells.
5. Infect T cells (1 wks)
6. Latently infected T cell → Tumor (3 wks)

7a. Tumor cells → nerves

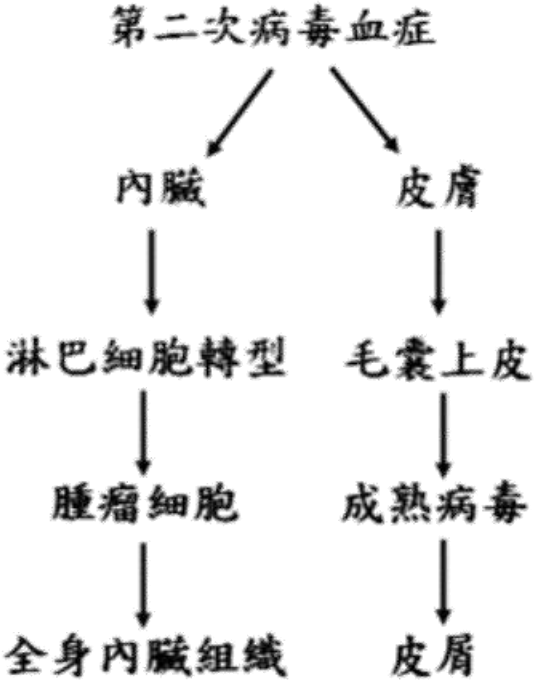
7b. Visceral lymphoma

8. Feather follicle → virus release into environment.

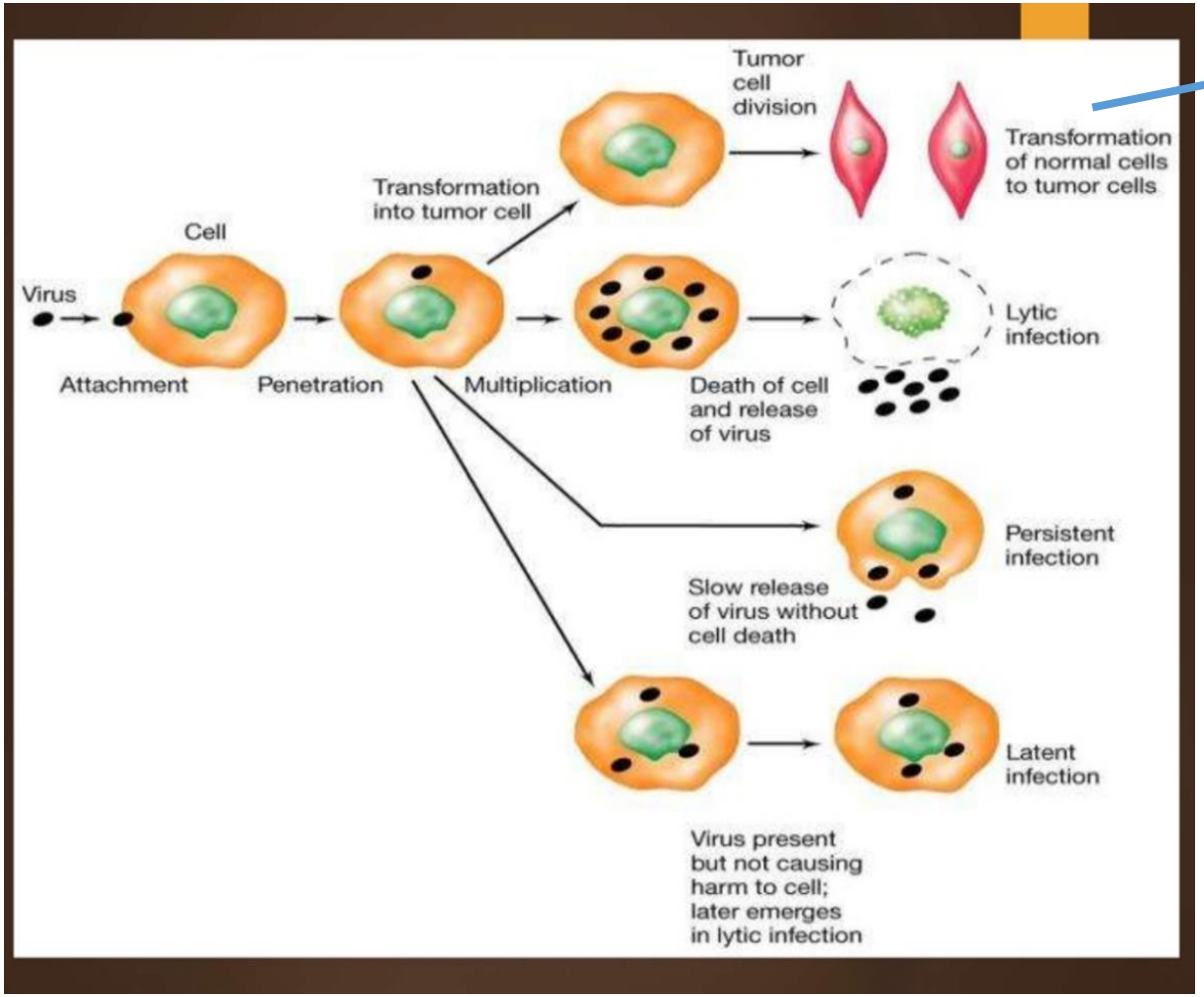
PATHOGENESIS







TRANSFORMATION OF LYMPHOCYTE



Lymphoma

DIAGNOSIS (OIE Terrestrial Manual 2018)

- clinical signs and gross or microscopic lesions
- Definitive diagnosis must be made by diagnosing the disease (tumor), not the infection.
- Chickens may become persistently infected with MDV without developing clinical disease.
- Infection by MDV is detected by virus isolation and the demonstration of viral nucleic acid, antigen or antibodies.

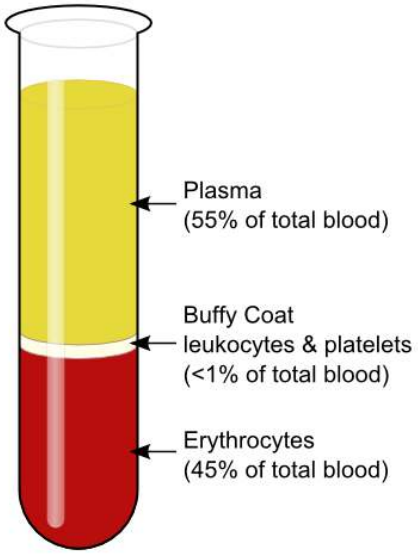
- Compared with the uniform cell populations observed in lymphoid leukosis, MD lymphomas consist of pleomorphic lymphoid cells of various types
- Vaccination greatly reduces clinical disease, but does not prevent persistent infection and shedding of MDV.

IDENTIFICATION OF THE AGENT

- virus isolation

buffy coat or tissue → chicken kidney cells (CK) or chicken/duck embryo fibroblasts (CEF/DEF) → few days: CPE or Plaques

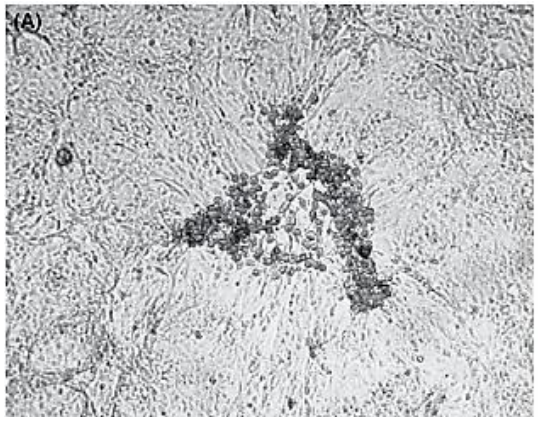
- PCR/qPCR
  - Tissue, Blood, feather tips
  - differentiating pathogenic and vaccine strains
  - ***Meq*** is the relevant oncogene

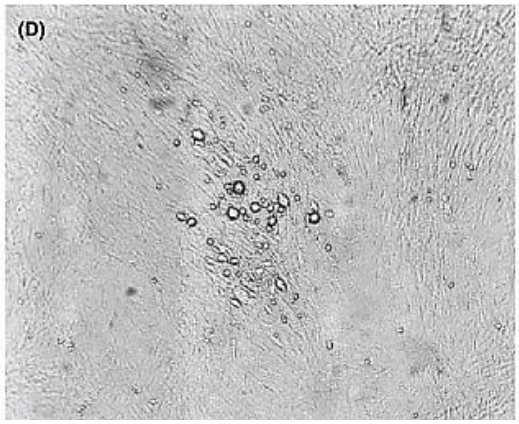


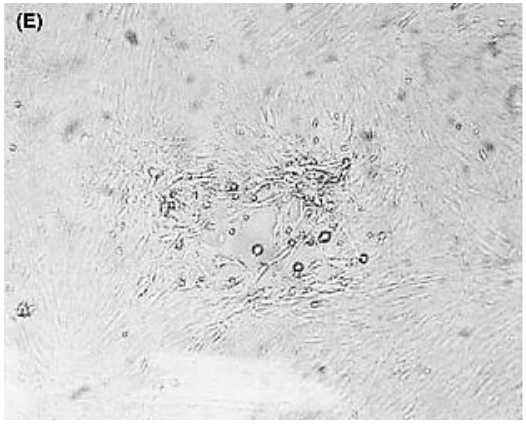
CHARACTERIZATION (CPE)

Serotype 1 | Serotype 2 | Serotype 3

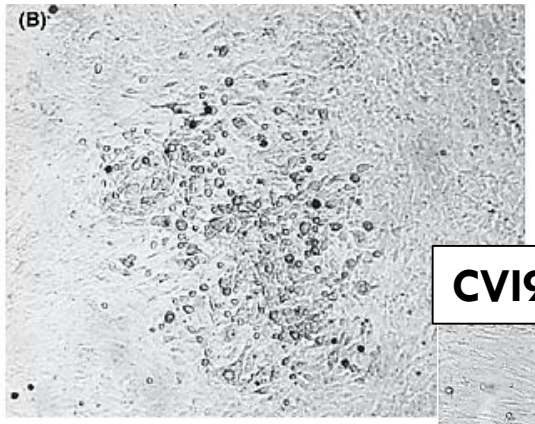
CEF
雞胚

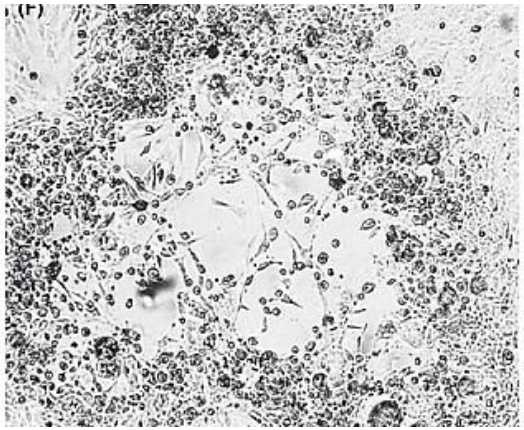




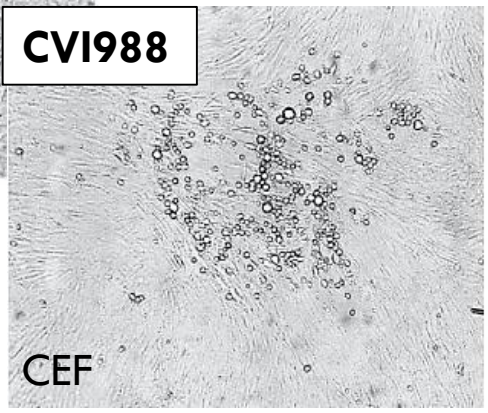


DEF
鴨胚



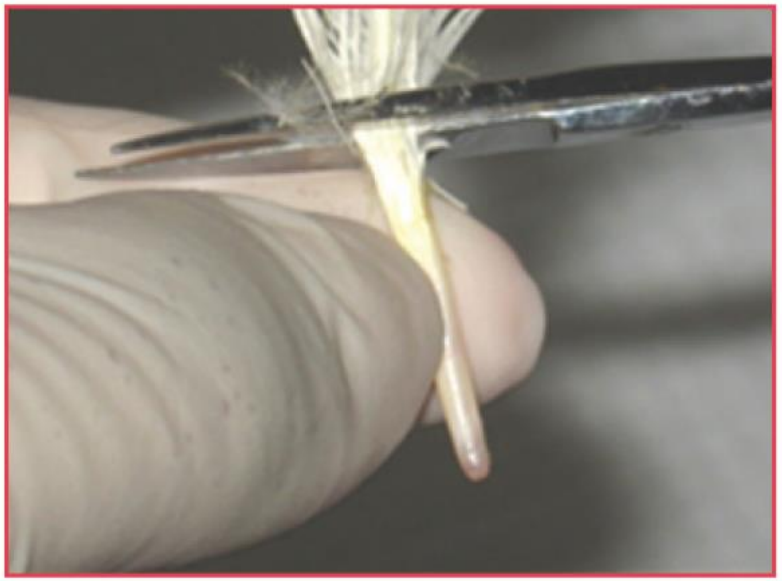


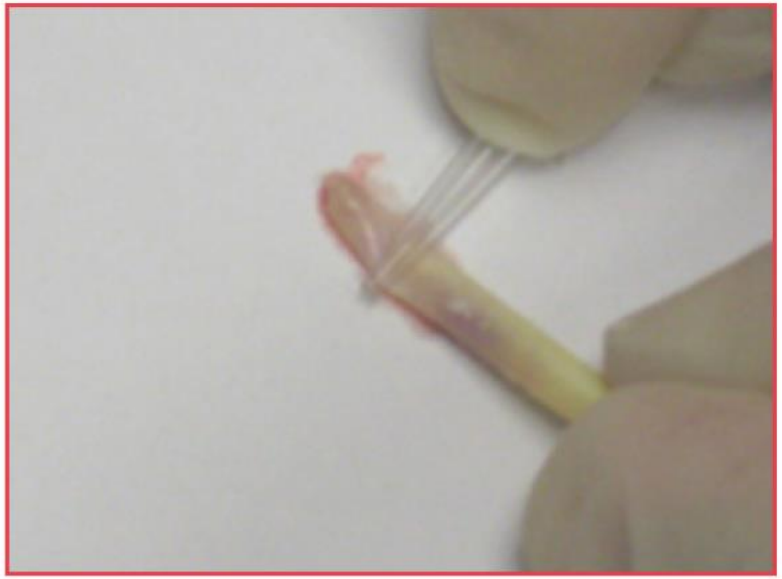
CVI988

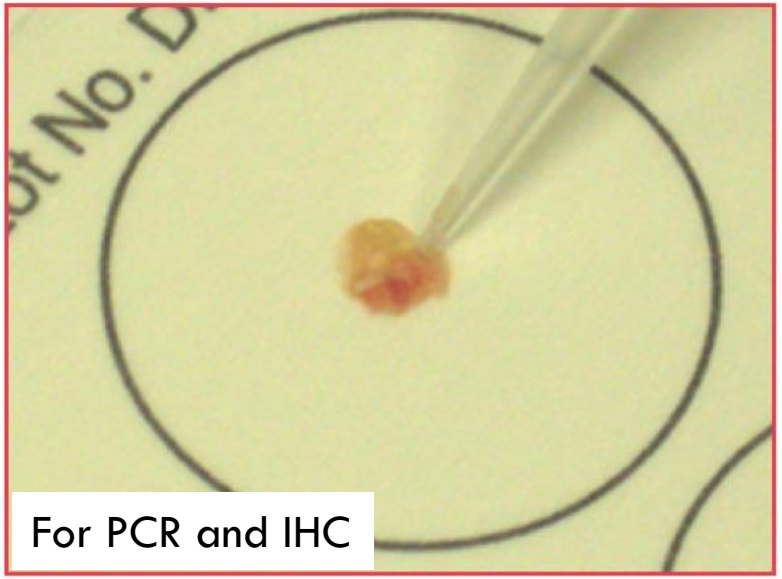


SAMPLE COLLECTION





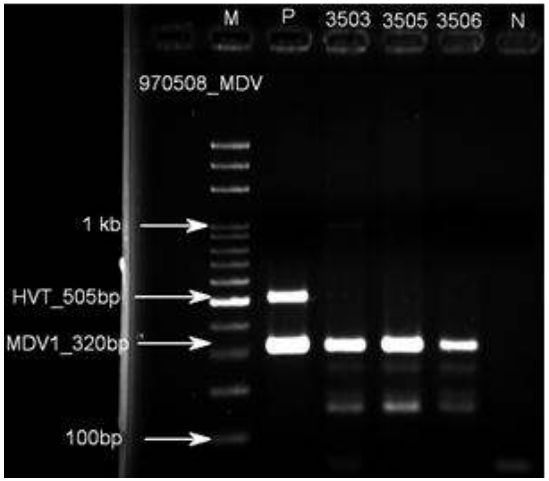




For PCR and IHC

CHARACTERIZATION (PCR)

- Virus DNA – PCR
- Primer sets
  - Vaccine/wildtype
  - *meq* gene
- Samples:
  - Tumor tissue
  - Feather follicle

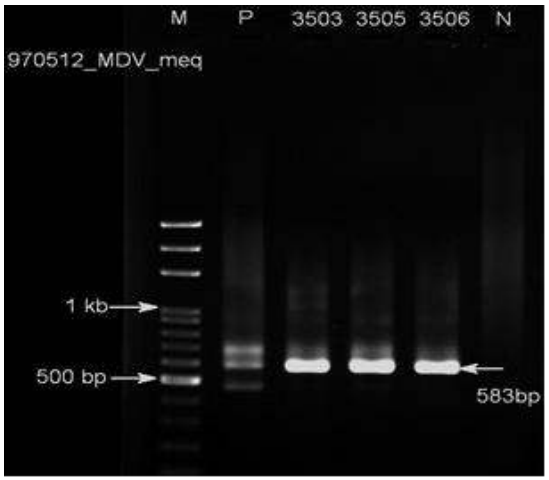


3503、3505、3506三檢體對馬立克病毒核酸MDV1測試呈陽性反應，產物大小約為320bp。

(註: HVT產物大小為505 bp)

M: 100 bp DNA marker;

P: positive control,
Lane 2-4, test samples;
N: negative control;
bp: base pairs



3503、3505、3506三檢體對馬立克病毒 *meq* 基因測試呈陽性反應，產物大小約為583 bp，為野外株。

(註: 疫苗株可見二條帶，大小約763 bp與583 bp)

M: 100bp DNA marker
P: positive control,
Lane 2-4, test samples;
N: negative control;
bp: base pairs

IFA (AGENT)

- MD : activated T cell antigens
- lymphoid leukosis : B-cell antigens

- these are not specific to MD tumour cells!

SEROLOGY

MDV infect chicken

- Neutralizing antibody in 1-2 weeks / IgM response → IgY response

- MDV is "**Cell-associated virus**" (細胞依附性病毒)

→ Neutralizing antibody only block cell-free virus
(early stage/environment)

- agar gel immunodiffusion test, or the indirect fluorescent antibody test

VACCINE

- 世界上第一個可用來預防腫瘤的疫苗

- Neutralizing antibody against “Cell-free virus”

- protect / against early replication, do not prevent infection

Type1	Type2	Type3
CVI988/Rispens	SB-1	HVT

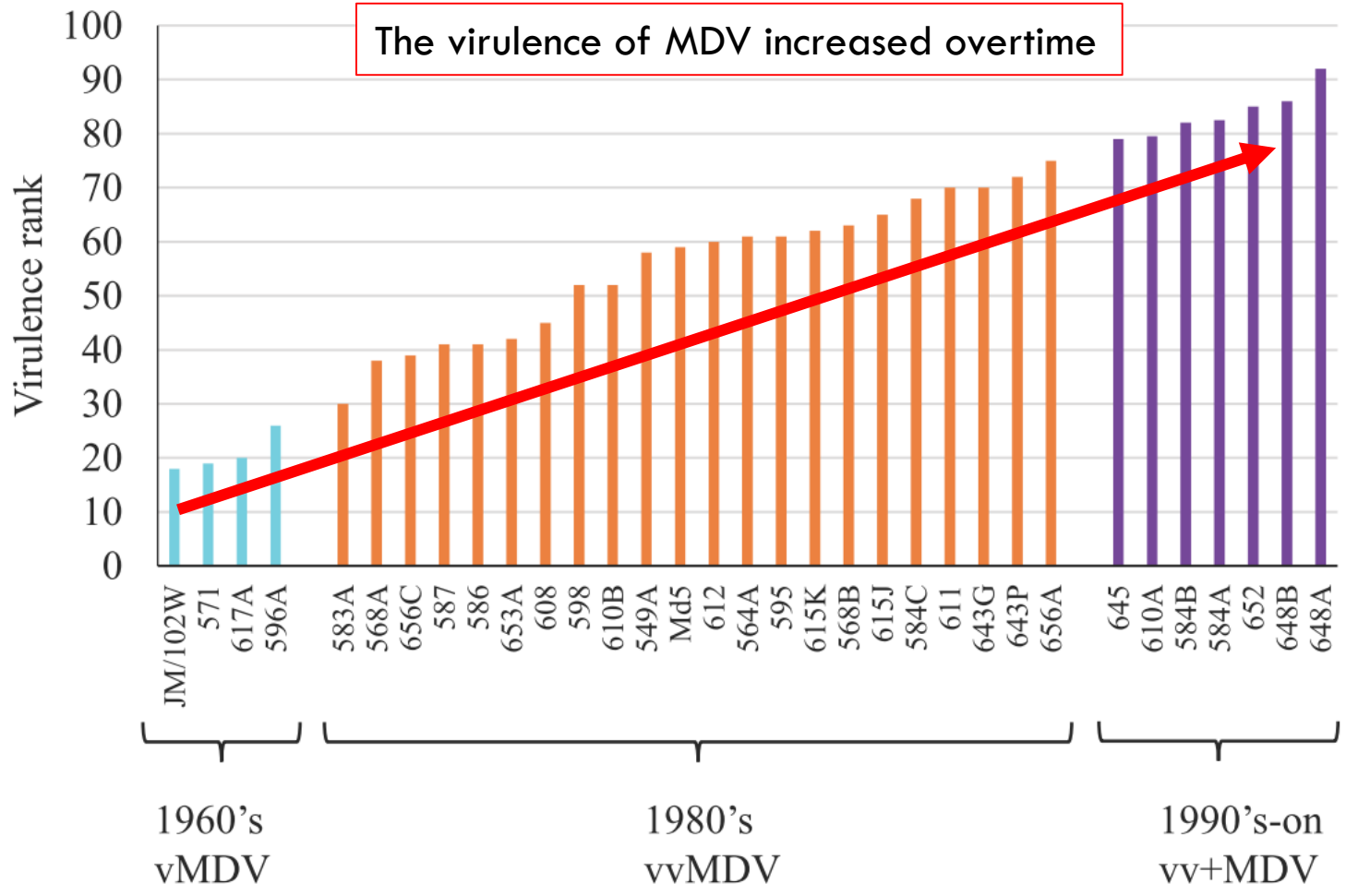
- common:

- **HVT** (Turkey) → for vMDV
- HVT + SB-1/CVI988 → for vvMDV (partial)
- HVT + CVI988 + SB-1 → for vvMDV (partial)

-in ovo vaccination (17-18day ovo)

VACCINE - CHALLENGE

Figure 1a: Evolution of MDV.



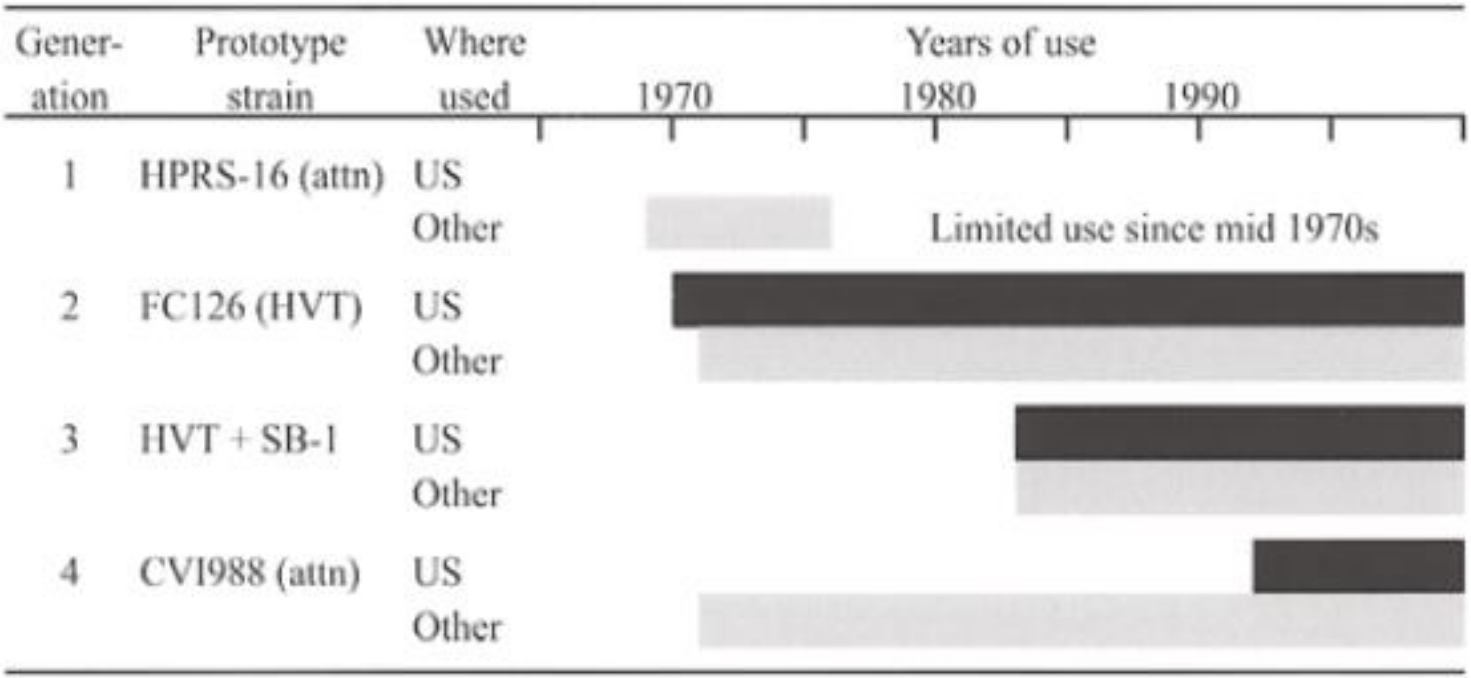
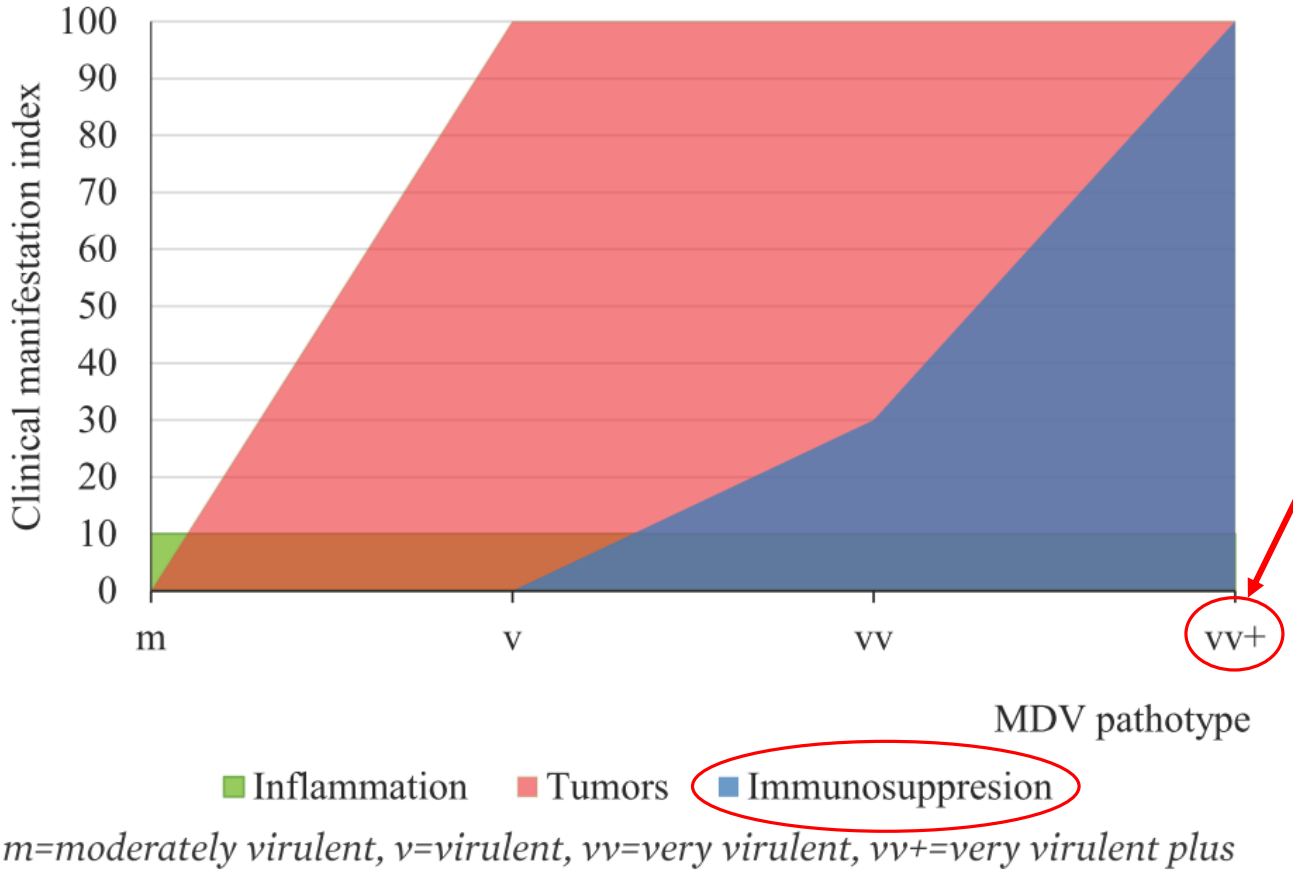


Fig. 2. Usage patterns of MD vaccines in the United States and other countries

VACCINE - CHALLENGE

Figure 1b: Consequences of MDV evolution for MD.



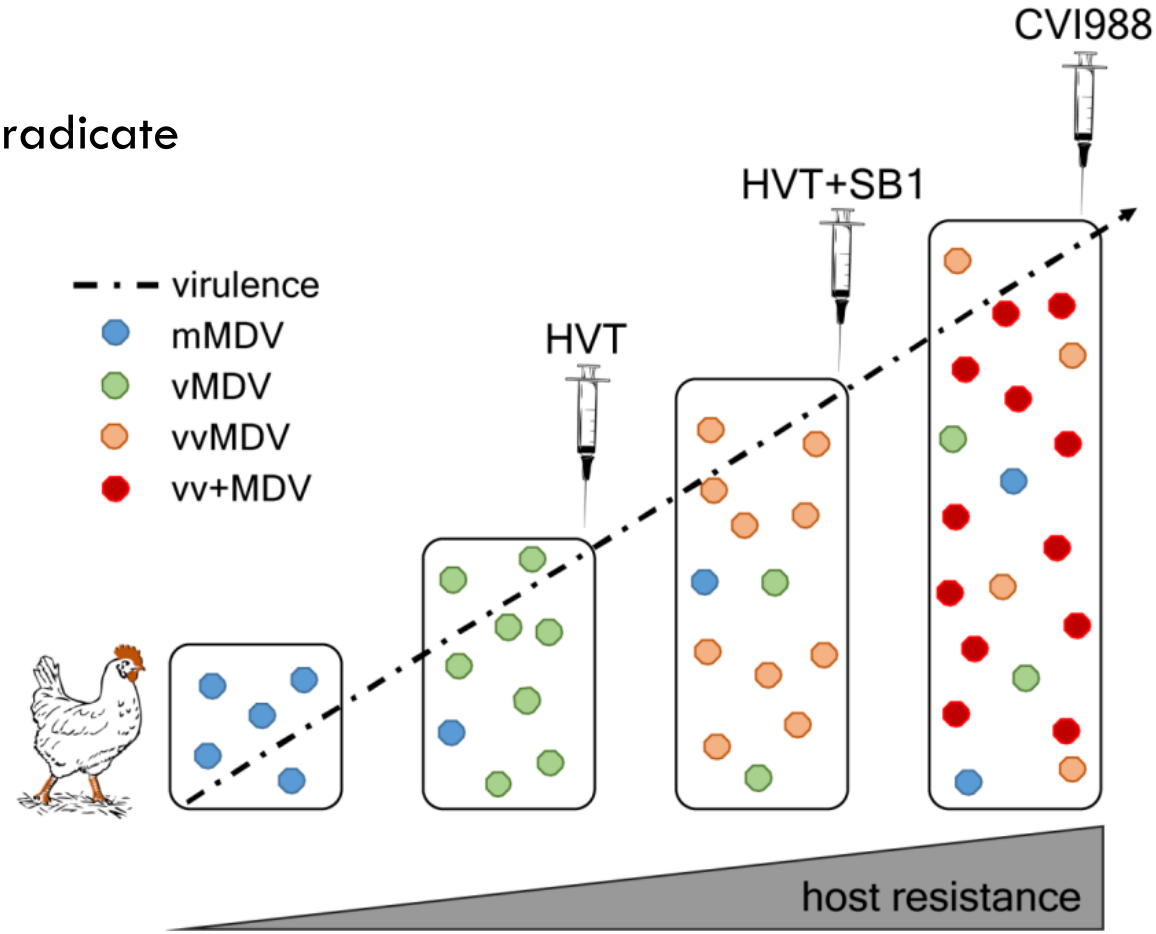
m=moderately virulent, v=virulent, vv=very virulent, vv+=very virulent plus

Acute cytolytic form!

- Die in 10-14day
- Atrophy of lymphoid

VACCINE - CHALLENGE

1. DNA virus = Stable
2. Environment = hard to eradicate
3. Cell-associated
4. Usage of vaccine



VACCINE - STORAGE

Storage





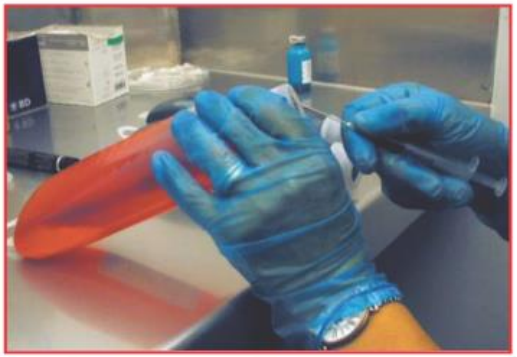
- Cell associated virus : HVT, SB-1, CVI988 or others
- Storage in -196 degree **liquid Nitrogen**
- cell-line: CEF/DEF … or others

VACCINE

Thawing
(water bath)



Reconstitution





Administration
(monitoring chicks)





MERIAL'S VAXXITEK & FROZEN MAREK VACCINES PREPARATION

1-EQUIPMENT



1 - The vaccine must be prepared in a clean and well ventilated room.



2 - Wear **protective** gloves sleeves and goggles when removing vaccine ampoules from liquid nitrogen



3 - Water-bath should contain a minimum of **5l** of clean water at a temperature of **27°C**. Water must be changed **daily**. For vaccine reconstitution use a **10ml** syringe and an **18G** needle.



4 - Ensure that diluent is at **room temperature (20-25°C)** and that the contents are translucent pink. Discard any bag of diluent if colour has changed from pink to yellowish or if it turns to cloudy.



5 - Recommandations !
Vaccination equipment must be **clean and disinfected.** Injection volume must be controlled **before each use.**

2-THAWING



1 - Mix only enough vaccine for **immediate use**.



2 - Thaw the ampoule by gently swirling it in waterbath. As soon as the ampoule is **totally thawed**, **remove immediately from the water bath**. Thaw time should not exceed **90 seconds.**

3-RECONSTITUTION



1 - **Swab the diluant bag port** with an alcohol wipe and allow to dry.

2 - **Wipe** the ampoule with a clean paper towel, **gently tap the tip** to recover of all the vaccine, and **open it immediately**.



3 - Slowly **remove the contents of the ampoule** by using a syringe fitted with an 18G needle.



4 - Slowly transfer the vaccine into the bag of diluent & **rinse the ampoule** and the tip with diluent. The vaccine should be added to the diluent as soon as it has been thawed.



5 - **Gently swirl the diluent bag to mix the vaccine**. **Do <u>not</u> shake vigorously**.



6 - **Note preparation time** on the bag of diluent once mixed. During vaccination, gently swirl the bag frequently to ensure a constant vaccine mix. **Once mixed, all vaccine must be used within one hour.**

RCS Lyon 570 800 215 - Illustrations : Vincent Hèye

Further For you

MERIAL

www.merial.com

OTHER VIRAL NEOPLASM DISEASES

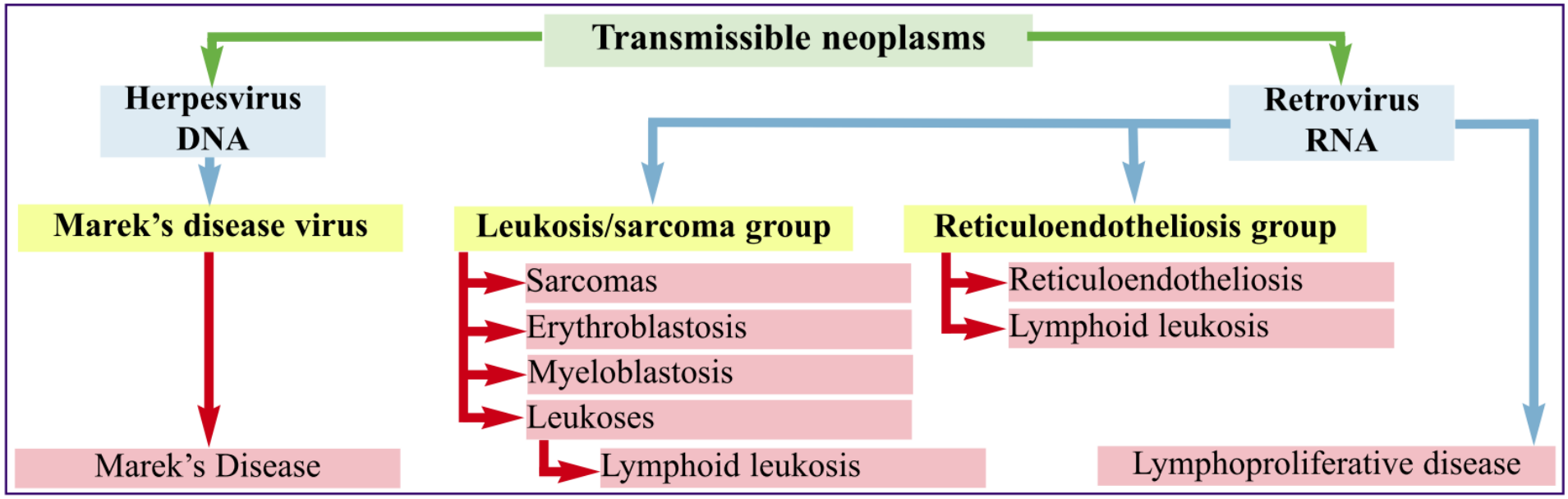


Table 1. *Features useful in differentiating Marek's disease, lymphoid leukosis and reticuloendotheliosis*

Feature	Marek's disease	Lymphoid leukosis	Reticuloendotheliosis*
Age	Any age. Usually 6 weeks or older	Not under 16 weeks	Not under 16 weeks
Signs	Frequently paralysis	Non-specific	Non-specific
Incidence	Frequently above 5% in unvaccinated flocks. Rare in vaccinated flocks	Rarely above 5%	Rare
Macroscopic lesions			
Neural involvement	Frequent	Absent	Infrequent
Bursa of Fabricius	Diffuse enlargement or atrophy	Nodular tumours	Nodular tumours
Tumours in skin, muscle and proventriculus, 'grey eye'	May be present	Usually absent	Usually absent
Microscopic lesions			
Neural involvement	Yes	No	Infrequent
Liver tumours	Often perivascular	Focal or diffuse	Focal
Spleen	Diffuse	Often focal	Focal or diffuse
Bursa of Fabricius	Interfollicular tumour and/or atrophy of follicles	Intrafollicular tumour	Intrafollicular tumour
Central nervous system	Yes	No	No
Lymphoid proliferation in skin and feather follicles	Yes	No	No

DIFFERENTIAL DIAGNOSIS

Table 3. Differential diagnosis of Marek's disease with other tumor and non-tumor diseases of poultry.

Age (weeks)	Gross lesions	Differential Diagnosis	Test to confirm diagnosis	
			Aid in the diagnosis	Confirm diagnosis
<14 MD -early	Enlarged nerves	Marek's disease Peripheral neuropathy	Histopathology	Real time PCR
	Tumors in viscera +Enlarged nerves	Marek's disease	Not necessary in most cases	Not necessary in most cases
	Tumors in viscera	Marek's disease Acutely transformed retrovirus	Histopathology Immunohistopathology	Real time PCR
>14 ALV/REV -late	Enlarged nerves	Marek's disease Peripheral neuropathy Reticuloendotheliosis (neuritis)	Histopathology	Real time PCR
	Tumors in viscera + Enlarged nerves	Marek's disease REV-induced tumors	Histopathology Immunohistochemistry	Real time PCR
	Tumors in viscera	Marek's disease ALV-induced tumors REV-induced tumors Spontaneous tumors	Histopathology Immunohistochemistry	Real time PCR
	Tumors in the Bursa of Fabricius	Marek's disease ALV-induced tumors REV-induced tumors	Histopathology Immunohistochemistry	Real time PCR

MDV IN TAIWAN



黃金城表示，臺灣總共進口七個廠牌的馬立克疫苗，進口時皆經過國家檢驗合格，但後續如儲存不良，可能造成疫苗效力下降。

- Controlled by vaccine
  >20 years

- Personalized vaccination program (!!!)
  - Combination of HVT + CVI988 or SB-1
  - Time frame for vaccination
  - 一般在孵化場施打再運送到飼養場所
  - Immunosuppress → influence **IBD/ND**

- 疫苗選擇應根據場內田間病毒株的強弱、污染的程度以及預期感染田間病毒的時間而決定。
- 當單價疫苗保護效果不足時，可採用雙價疫苗。

預防與控制

正確/適當的接種技術、注射時劑量要足

嚴格的生物防護，統進統出

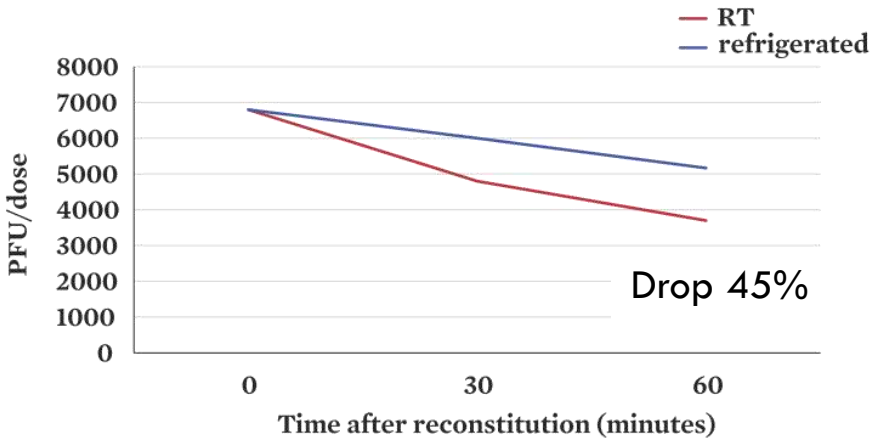
移除使用過的墊料，徹底清潔消毒

降低緊迫因素如熱、高度潮溼、過多的氨及缺乏食物及水。

基因的選擇增進雞隻對馬立克病之抵抗力

避免各種免疫抑制疾病發生

使用與種雞不同的疫苗



Room temperature: reduced to 55% within 1h
Refrigeration: reduced to 76% within 1h

MD疫苗失敗的原因

- 疫苗有效力價不足
- 雞場衛生管理或生物安全不確實
- 移行抗體干擾
- 免疫抑制的影響
- 早期接觸到野外毒
- 添加抗生素對疫苗效價的影響，不可使用 Aureomycin、Terramycin、Erythromycin、Tetracyeline 等
- 馬立克疫苗防疫失敗原因複雜，除野外毒的毒力變異外，疫苗品質、接種技術、禽場生物安全以及飼養管理皆可能牽涉在內，宜加強相關工作人員對疫苗的冷鏈運輸、貯存、配製及正確的施打程序之訓練。

SUMMARY

- 淋巴球增生性疾病、有多種病型
- 吸入毛囊細胞、皮屑而水平傳播
- 以病毒學技術、病理學檢查共同診斷
- 需與其他腫瘤疾病(AL, RE) 區別
- 以細胞型疫苗預防 (須注意疫苗保存)
- MDV為重要的免疫抑制性病原，高風險商業雞場應加強生物安全措施與飼養管理，以避免其他病原合併及繼發性感染